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# GeoEvo: Identity-Aware Potential Game with Geometric Evolution for Personalized Multimodal Federated Learning

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## Abstract

We reconceptualize Personalized Multimodal Federated Learning (PMFL) by treating missing modalities as intrinsic structural identities that constrain each client to a distinct Riemannian submanifold, rather than as deficiencies to be compensated. To reconcile the tension between identity preservation and cross-client collaboration, we cast PMFL as an identity-aware potential game and seek a geometry-consistent equilibrium instead of a monolithic full-modality optimum. We propose GeoEvo, a federated approximate solver that combines curvature-adaptive Fisher descent with manifold-lifted evolutionary search: Natural Evolution Strategies for basin escape and Particle Swarm updates anchored to a server-broadcast Fréchet prototype for cross-client transfer. A monotone selection rule guarantees surrogate descent and an  $\mathcal{O}(1/\sqrt{T})$  stationarity rate; controlled lift residuals further imply expected potential dissipation and, when vanishing, convergence toward first-order Nash equilibria. Empirically, GeoEvo improves personalization and robustness across diverse missing-modality patterns.

## 1. Introduction

Federated Learning (FL) (McMahan et al., 2017) has emerged as a promising paradigm for collaborative model training across distributed clients without exposing raw data. However, conventional FL (Li et al., 2020) remains predominantly unimodal despite real-world multimodality. To bridge this gap, Multimodal Federated Learning

(MMFL) (Xiong et al., 2022) extends FL to exploit cross-modal synergies, learning richer multimodal representations while preserving strict privacy guarantees.

Existing MMFL frameworks (Sun et al., 2024; Singha et al., 2025) presume full modality availability, treating clients as homogeneous entities that differ only in data distribution. In practice, this presupposition collapses under heterogeneous hardware, divergent privacy constraints, or asymmetric sensors. The resulting modality incompleteness shifts the core challenge from managing statistical discrepancies to coordinating a fragmented federation with structurally incompatible observation domains. Such fragmentation creates a persistent performance gap; although Personalized Federated Learning (PFL) (Tan et al., 2022a) seeks to address client-specific characteristics, it remains constrained by a shared input space. Unlike standard statistical shifts, this incompleteness introduces *structural heterogeneity*—a more profound barrier where divergent hypothesis-space dimensions render conventional personalization ill-posed.

Prevailing approaches (Yu et al., 2023; Nguyen et al., 2026) treat missing modalities as deficiencies to be rectified, relying on imputation or robust training to enforce alignment with a modality-complete reference. This compensation-centric view, however, conflates structural voids with distributional shifts, pursuing correspondence where none exists. We argue that a client’s modality configuration constitutes its *intrinsic structural identity*—not a defect, but a defining characteristic constraining optimization to a distinct Riemannian submanifold. This conceptual shift motivates **Personalized Multimodal Federated Learning (PMFL)**, which realizes cross-client collaboration via geometry-aware adaptation across heterogeneous manifolds, circumventing representational distortions of forced global alignment.

This perspective exposes a coordination paradox: *how can clients on distinct manifolds collaborate without collapsing into an identity-erasing consensus?* We resolve this tension by casting PMFL as an **Identity-Aware Potential Game**, where each client optimizes on its Riemannian manifold, coupled through a shared exact potential. This formulation reduces decentralized objective conflicts to principled descent on a potential landscape, reconciling collaborative alignment with preservation of heterogeneous equilibria.

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Exact minimization of potential is hindered by structural barriers: *communication infeasibility*, as pairwise couplings require neighbor synchronization; and *multimodal entrapment*, as non-convex potential confines first-order descent to disconnected basins. We propose **GeoEvo** (Geometric Evolution), a federated approximate solver dissolving both via complementary server-client operations. On the server, Fréchet prototypes compress the  $\mathcal{O}(K^2)$  pairwise couplings into one message per client. On each client, a Fisher-preconditioned retracted update adapts descent to identity-induced curvature, while a manifold-lifted evolutionary search—combining Natural Evolution Strategies (NES) for stochastic basin escape with Particle Swarm Optimization (PSO) anchored to the broadcast prototype—explores beyond the local basin. A monotone selection rule guarantees per-step surrogate descent and, under controlled lift residuals, expected global-potential dissipation.

Our contributions are threefold: (i) **Conceptual Reframing**. We reconceptualize PMFL by treating modality incompleteness as an intrinsic structural identity that constrains each client to a distinct Riemannian submanifold, shifting the paradigm from compensatory alignment to geometry-aware adaptation. (ii) **Game-Theoretic Foundation**. We formulate PMFL as an Identity-Aware Potential Game, reducing decentralized multi-objective coordination to principled descent on a scalar potential landscape with guaranteed convergence to geometry-consistent first-order Nash equilibria. (iii) **Algorithmic Realization**. We develop GeoEvo, a federated solver that pairs curvature-adaptive Fisher descent with manifold-lifted evolutionary search. Server-broadcast Fréchet prototypes reduce communication to  $\mathcal{O}(K)$  messages per round; a monotone selection rule guarantees surrogate descent and  $\mathcal{O}(1/\sqrt{T})$  stationarity rate, with expected global-potential dissipation under controlled lift residuals.

## 2. Problem Formulation

### 2.1. Personalized Federated Learning

Standard FL seeks a single global model  $\theta_g$  by minimizing the weighted empirical risk across  $K$  clients:  $\theta_g = \arg \min_{\theta} \sum_{k=1}^K w_k F_k(\theta)$ , where  $F_k(\theta) = \mathbb{E}_{(x,y) \sim \mathcal{D}_k} [\ell(f_{\theta}(x), y)]$  is the local objective,  $\mathcal{D}_k$  denotes the local data distribution, and  $w_k \geq 0$  is the aggregation weight with  $\sum_{k=1}^K w_k = 1$ . Under statistical heterogeneity, a universal  $\theta_g$  generally fails to attain each client’s local optimum  $\theta_k^*$ , incurring a nonnegative optimality gap  $F_k(\theta_g) - F_k(\theta_k^*) \geq 0$ , which is strict whenever  $\theta_g \notin \arg \min_{\theta} F_k(\theta)$ . To mitigate this, PFL aims to learn client-specific models  $\{\theta_k\}_{k=1}^K$  by jointly leveraging collaborative training and local distribution adaptation, a problem commonly cast as multi-task optimization:  $\min_{\{\theta_k\}_{k=1}^K} \sum_{k=1}^K w_k F_k(\theta_k)$ . A classical solution introduces a proximal regularizer (Li et al., 2021):  $\min_{\theta_k} F_k(\theta_k) + \frac{\lambda}{2} \|\theta_k - \theta_g\|^2$ .

However, existing PFL methods implicitly assume (i) homogeneous client parameterization and (ii) a shared input space  $\mathcal{X}$ . Both assumptions break down in multimodal settings, where clients operate on distinct feature spaces and often demand heterogeneous architectures.

### 2.2. Multimodal Federated Learning

The collapse of standard PFL premises in multimodal scenarios reveals a deeper barrier: *structural heterogeneity*, as clients observe data from divergent modality subsets over incompatible domains. Let  $\mathcal{M} = \{m_1, \dots, m_M\}$  denote the universe of modalities, with idealized full input space  $\mathcal{X} = \prod_{m \in \mathcal{M}} \mathcal{X}^{(m)}$ . Each client  $k$  observes only  $\mathcal{M}_k \subseteq \mathcal{M}$ , inducing a client-specific input space  $\mathcal{X}_k = \prod_{m \in \mathcal{M}_k} \mathcal{X}^{(m)}$ . Given this client-specific input  $\mathbf{x}_{\mathcal{M}_k} \in \mathcal{X}_k$ , the standard MMFL objective is  $\theta_g = \arg \min_{\theta} \sum_{k=1}^K w_k F_k(\theta; \mathcal{M}_k)$ , where  $F_k(\theta; \mathcal{M}_k) = \mathbb{E}_{(\mathbf{x}_{\mathcal{M}_k}, y) \sim \mathcal{D}_k} [\ell(f_{\theta}(\mathbf{x}_{\mathcal{M}_k}), y)]$ , exposing the structural mismatch: when  $\mathcal{M}_i \neq \mathcal{M}_j$ , the optimization landscapes are defined on dimensionally incompatible product spaces. Consequently, global parameter averaging becomes semantically ill-posed, as no canonical bijection exists between disparate sensor configurations that would justify a shared Euclidean coordinate system.

**Optimization Divergence.** This structural mismatch exacerbates gradient heterogeneity, quantified by  $\Gamma(\theta) = \sum_{k=1}^K w_k \|\nabla_{\theta} F_k(\theta; \mathcal{M}_k) - \nabla_{\theta} \bar{F}(\theta)\|^2$ , with  $\bar{F}(\theta) = \sum_{k=1}^K w_k F_k(\theta; \mathcal{M}_k)$ . Unlike unimodal FL where  $\Gamma$  reflects purely distributional shifts  $\mathcal{D}_i \neq \mathcal{D}_j$  (Karimireddy et al., 2020), in MMFL,  $\Gamma$  captures structural incompatibility: even with identical distributions, gradients inhabit different effective subspaces, pointing to irreconcilable optima.

**Proposition 2.1** (Structural Divergence of Local Optima). *Let clients  $i$  and  $j$  share the same underlying full-modality distribution, i.e.,  $\mathcal{D}_i|_{\mathcal{M}} = \mathcal{D}_j|_{\mathcal{M}}$ , but observe different modality subsets  $\mathcal{M}_i \neq \mathcal{M}_j$ . Assume that distinct modality-conditioned Bayes-optimal predictors induce disjoint population-risk minimizer sets under the shared parameterization. If the Bayes-optimal predictors associated with  $\mathcal{M}_i$  and  $\mathcal{M}_j$  differ on a set of positive probability, then  $\arg \min_{\theta \in \Theta} F_i(\theta; \mathcal{M}_i) \neq \arg \min_{\theta \in \Theta} F_j(\theta; \mathcal{M}_j)$ .*

Under a strictly proper loss, the optimal predictor is the conditional Bayes act associated with client’s observed  $\sigma$ -field. Distinct conditional Bayes acts yield distinct parameter-space optima under the stated identifiability condition.

### 2.3. Limitations of Vanilla Extensions

We first formalize a vanilla extension of PFL to the multimodal domain as our baseline:

$$\min_{\theta_g, \{\theta_k\}} \sum_{k=1}^K w_k F_k(\theta_k; \mathcal{M}_k) + \frac{\lambda}{2} \sum_{k=1}^K \|\theta_k - \theta_g\|^2. \quad (1)$$

Eq. (1) admits client-specific parameters  $\theta_k$ , yet treats modality absence as a statistical deficiency of  $\mathcal{D}_k$  rather than a structural property of client  $k$  itself.

**Definition 2.2** (Static Absence as Structural Identity). For client  $k$ , static absence denotes the permanent unavailability of modalities outside its fixed configuration  $\mathcal{M}_k$ , imposed by time-invariant physical constraints and distinct from transient dynamic occlusion. We refer to  $\mathcal{M}_k$  as client  $k$ 's *structural identity*: an invariant of the observation space  $\mathcal{X}_k$  that is irreducible to a statistical deficiency of  $\mathcal{D}_k$ .

**Irreducible Optimality Gap.** To compare population risks across modality subsets, we assume an underlying full-modality distribution  $\mathcal{D}_k|_{\mathcal{M}}$  whose marginalization onto  $\mathcal{M}_k$  recovers  $\mathcal{D}_k$ ; for any  $\mathcal{M}' \subseteq \mathcal{M}$ , define  $F_k(\theta; \mathcal{M}') \triangleq \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}_k|_{\mathcal{M}'}}[\ell(f_\theta(\mathbf{x}_{\mathcal{M}'}, y))]$  and  $F_k^*(\mathcal{M}') \triangleq \min_{\theta \in \Theta} F_k(\theta; \mathcal{M}')$ . Since restricting predictors to the smaller  $\sigma$ -field  $\sigma(\mathcal{X}_{\mathcal{M}_k})$  cannot decrease the Bayes risk,  $F_k^*(\mathcal{M}_k) \geq F_k^*(\mathcal{M})$ , with strict inequality whenever the omitted modalities carry information about  $Y$  not contained in  $\mathcal{X}_k$ . In such cases, richer-modality clients pull  $\theta_g$  toward full-modality minimizers, while a modality-limited client cannot attain the risk  $F_k^*(\mathcal{M})$ , forcing it to forfeit personalization for an unreachable consensus.

**Gradient Misalignment.** Beyond mis-targeting the optimum, Euclidean coupling injects unidentifiable directions.

**Proposition 2.3** (Incompatible Update Directions under Structural Mismatch). *Under the nested block-Fisher support condition formalized in Assumption B.6, let  $\mathcal{M}_i \subsetneq \mathcal{M}_j$ , and let  $S_k(\theta) \subseteq \mathbb{R}^d$  denote the identifiable subspace of client  $k$  at  $\theta$ , defined as the range of its local Fisher information. Then  $S_i(\theta) \subsetneq S_j(\theta)$ ,  $S_j(\theta) \cap S_i(\theta)^\perp \neq \{0\}$ . If  $P_{S_i(\theta)^\perp}(\theta_g - \theta_i) \neq 0$ , then the proximal pull in Eq. (1) contains a locally unidentifiable component that produces no first-order decrease of  $F_i(\theta; \mathcal{M}_i)$  and may cause wasted regularization motion or negative transfer.*

**Critique of the Robustness Paradigm.** A natural alternative is worst-case robust optimization over modality subsets:  $\min_{\theta \in \Theta} \max_{\mathcal{M}' \subseteq \mathcal{M}} F_k(\theta; \mathcal{M}')$ . While appropriate for dynamic modality failures, where any subset may manifest at test time, this objective is mis-specified under static constraints: it conflates transient corruption with the client's permanent structural identity, sacrificing personalization for robustness against absence patterns that, by Definition 2.2, never materialize for that client.

## 2.4. PMFL: Identity-Aware Potential Game

Motivated by the structural failures of Eq. (1), we reformulate PMFL as a *potential game* (Monderer & Shapley, 1996) over identity-constrained Riemannian manifolds (Ab-sil et al., 2008), in which each client optimizes within its feasible geometry while contributing to a common global

potential that encodes only structurally compatible transfer.

**Definition 2.4** (Client Identity and Feasible Geometry). Extending the structural identity  $\mathcal{M}_k$  of Definition 2.2 with the local distribution, we define the *client identity* as the pair  $\mathcal{I}_k \triangleq (\mathcal{D}_k, \mathcal{M}_k)$ . This identity restricts client  $k$ 's local optimization to a feasible geometry  $\Theta_k \subset \mathbb{R}^{d_k}$ , assumed to be a finite-dimensional smooth embedded manifold admitting a second-order retraction, endowed with a metric  $g_k$  intrinsically determined by  $\mathcal{I}_k$ .

**Definition 2.5** (Transferable Structure Space). To enable principled comparison between heterogeneous feasible geometries, let  $(\mathcal{Z}, d_{\mathcal{Z}})$  be a common metric space, the *transferable structure space*. For each client  $k$ , an extraction map  $\mathcal{T}_k : \Theta_k \rightarrow \mathcal{Z}$  maps local parameters to structural representation  $z_k \triangleq \mathcal{T}_k(\theta_k)$ . For any modality subset  $\mathcal{M}' \subseteq \mathcal{M}$ , a restriction map  $\pi_{\mathcal{M}'} : \mathcal{Z} \rightarrow \mathcal{Z}_{\mathcal{M}'}$  projects onto the subspace supported on  $\mathcal{M}'$ , with  $d_{\mathcal{Z}}$  inducing a metric on each  $\mathcal{Z}_{\mathcal{M}'}$  by restriction.

**Assumption 2.6** (Structural Compatibility). We assume:

(i) **Regularity:** For each client  $k$ ,  $\mathcal{T}_k$  is measurable and locally Lipschitz on  $\Theta_k$ . (ii) **Subspace Comparability.** For any  $\mathcal{M}' \subseteq \mathcal{M}$ ,  $\pi_{\mathcal{M}'}$  is well-defined. For any pair  $(i, j)$  with  $S_{ij} \triangleq \mathcal{M}_i \cap \mathcal{M}_j \neq \emptyset$ ,  $\pi_{S_{ij}}(z_i)$  and  $\pi_{S_{ij}}(z_j)$  lie in the same metric space  $(\mathcal{Z}_{S_{ij}}, d_{\mathcal{Z}})$  for all  $\theta_i \in \Theta_i$ ,  $\theta_j \in \Theta_j$ . (iii) **Sparse Interaction:** Symmetric couplings satisfy  $\omega_{ij} = \omega_{ji} \geq 0$ , with  $\omega_{ij} > 0$  if and only if  $S_{ij} \neq \emptyset$ .

**Game Formulation.** Let  $W \triangleq (\theta_1, \dots, \theta_K) \in \prod_{k=1}^K \Theta_k$  denote the joint strategy profile and  $W_{-k}$  the profile with  $\theta_k$  excluded. We formulate PMFL as a  $K$ -player non-cooperative game where each client  $k$  minimizes a payoff combining local fitness with pairwise structural alignment:

$$J_k(\theta_k; W_{-k}) \triangleq w_k F_k(\theta_k; \mathcal{I}_k) + \frac{\lambda}{2} \sum_{j \in \mathcal{N}_k} \omega_{kj} d_{\mathcal{Z}}^2(\pi_{S_{kj}}(z_k), \pi_{S_{kj}}(z_j)), \quad (2)$$

where  $\mathcal{N}_k \triangleq \{j \in [K] \setminus \{k\} : S_{kj} \neq \emptyset\}$  collects  $k$ 's structurally compatible neighbors. A profile  $W^*$  is a local Nash equilibrium if, for every  $k \in [K]$ , there exists a neighborhood  $U_k$  of  $\theta_k^*$  such that  $\theta_k^* \in \arg \min_{\theta \in U_k \cap \Theta_k} J_k(\theta; W_{-k}^*)$ .

**Exact Potential Function.** Computing Nash equilibria is PPAD-hard (Daskalakis et al., 2006); the symmetric coupling  $\omega_{ij} = \omega_{ji}$  in Eq. (2), by contrast, collapses these per-client neighbor sums into a single exact potential  $V : \prod_{k=1}^K \Theta_k \rightarrow \mathbb{R}$ , counting each unordered pair  $\{i, j\}$  once:

$$V(W) \triangleq \sum_{k=1}^K w_k F_k(\theta_k; \mathcal{I}_k) + \frac{\lambda}{2} \sum_{1 \leq i < j \leq K} \omega_{ij} d_{\mathcal{Z}}^2(\pi_{S_{ij}}(z_i), \pi_{S_{ij}}(z_j)). \quad (3)$$

**Proposition 2.7** (Exact Potential Game). *Under Assumption 2.6(iii),  $V(W)$  in Eq. (3) is an exact potential for the game in Eq. (2): for any client  $k$ , profile  $W_{-k}$ , and unilateral deviation  $\theta'_k \in \Theta_k$ ,*

$$J_k(\theta'_k; W_{-k}) - J_k(\theta_k; W_{-k}) = V(\theta'_k, W_{-k}) - V(\theta_k, W_{-k}). \quad (4)$$

*Consequently, every local minimizer of  $V$  on  $\prod_{k=1}^K \Theta_k$  is a local Nash equilibrium of the game.*

**Stability via Riemannian Gradient Flow.** Beyond showing that local minimizers of  $V$  are local Nash equilibria, the potential structure yields a clean stability analysis via gradient flow. On the product manifold  $\mathcal{W} \triangleq \prod_{k=1}^K \Theta_k$ , consider the continuous-time dynamics

$$\dot{\theta}_k(t) = -\text{grad}_{g_k, \theta_k} V(W(t)), \quad k = 1, \dots, K, \quad (5)$$

where  $\text{grad}_{g_k, \theta_k} V$  denotes the Riemannian gradient of  $V$  on  $\Theta_k$  under  $g_k$ . Along Eq. (5),  $V$  acts as a Lyapunov function:

$$\frac{d}{dt} V(W(t)) = - \sum_{k=1}^K \|\text{grad}_{g_k, \theta_k} V(W(t))\|_{g_k}^2 \leq 0, \quad (6)$$

Equality holds if and only if  $\text{grad}_{g_k, \theta_k} V(W(t)) = 0$  for all  $k$ , i.e.,  $W(t)$  is a first-order stationary point of  $V$ ; by Proposition 2.7, every player is also first-order stationary. If the initial sublevel set  $\mathcal{L}(V(W(0)))$  is compact, LaSalle's invariance principle implies that the trajectory approaches this stationary set.

**Decentralized Realization.** By Proposition 2.7,  $\text{grad}_{g_k, \theta_k} J_k(\theta_k; W_{-k}) = \text{grad}_{g_k, \theta_k} V(W)$  for all  $k \in [K]$ . Hence the gradient flow in Eq. (5) is equivalent to the decentralized dynamics

$$\dot{\theta}_k(t) = -\text{grad}_{g_k, \theta_k} J_k(\theta_k(t); W_{-k}(t)), \quad k = 1, \dots, K,$$

whereby each client independently descends its own payoff  $J_k$ , thereby monotonically decreasing the global potential  $V$  and driving  $W(t)$  toward the set of first-order Nash equilibria, without explicit coordination on  $V$ .

Full proofs and technical details are deferred to Appendix B.

### 3. Methodology

#### 3.1. Reference Solver: Exact Potential Minimization

By Proposition 2.7, solving the identity-aware PMFL game in Eq. (2) reduces to minimizing the exact potential  $V(W)$  over the product space  $\mathcal{W} = \prod_{k=1}^K \Theta_k$ :

$$\begin{aligned} \min_{W \in \mathcal{W}} V(W) &= \sum_{k=1}^K w_k F_k(\theta_k; \mathcal{I}_k) \\ &\quad + \frac{\lambda}{2} \sum_{1 \leq i < j \leq K} \omega_{ij} d_{\mathcal{Z}}^2(\pi_{S_{ij}}(z_i), \pi_{S_{ij}}(z_j)). \end{aligned} \quad (7)$$

**Identity-Consistent Fisher Metric.** The minimization landscape of  $V$  is highly anisotropic across clients: coordinates affecting frequently observed modalities differ in sensitivity from those affecting rare ones. Concretely, we realize  $\Theta_k$  (Definition 2.4) as a submanifold of an ambient modality-factorized parameterization in which the blocks of absent modalities  $\mathcal{M} \setminus \mathcal{M}_k$  are inactive. To equip each  $\Theta_k$  with an identity-aware metric  $g_k$ , we observe that, for such models, the score  $\nabla_{\theta_k} \log p(y | \mathbf{x}_{\mathcal{M}_k}; \theta_k)$  vanishes along these inactive coordinates. The empirical Fisher information matrix (FIM) (Kunstner et al., 2019) thereby inherits the block-sparse structure of  $\mathcal{M}_k$  without explicit masking:

$$\mathbf{F}_k(\theta_k) \triangleq \frac{1}{|\widehat{\mathcal{D}}_k|} \sum_{(\mathbf{x}, y) \in \widehat{\mathcal{D}}_k} (\nabla_{\theta_k} \log p(y | \mathbf{x}_{\mathcal{M}_k}; \theta_k)) (\nabla_{\theta_k} \log p(y | \mathbf{x}_{\mathcal{M}_k}; \theta_k))^\top, \quad (8)$$

where  $\widehat{\mathcal{D}}_k$  denotes the local sample set drawn from  $\mathcal{D}_k$  and  $p(y | \mathbf{x}_{\mathcal{M}_k}; \theta_k)$  is the predictive distribution induced by  $f_{\theta_k}$ .

**Remark 3.1** (Damped Natural-Gradient Preconditioning). For numerical stability, we use  $\widetilde{\mathbf{F}}_k(\theta_k) \triangleq \mathbf{F}_k(\theta_k) + \epsilon I$  with  $\epsilon > 0$ . Its restriction to  $T_{\theta_k} \Theta_k$  defines the metric  $g_k$ , and  $\widetilde{\mathbf{F}}_k(\theta_k)^{-1}$  serves as the natural-gradient preconditioner.

**Block Riemannian Descent.** By the decentralized realization of Section 2.4, each client can update independently while collectively minimizing  $V$ . We discretize the gradient flow in Eq. (5) via block-wise retracted steps:

$$\theta_k^{t+1} = \text{Retr}_{\theta_k^t}(-\eta_t \text{grad}_{g_k, \theta_k} V(W^t)), \quad k = 1, \dots, K, \quad (9)$$

where, in local coordinates, the Fisher metric gives  $\text{grad}_{g_k, \theta_k} V = \widetilde{\mathbf{F}}_k(\theta_k)^{-1} \nabla_{\theta_k} V(W^t)$ , with  $\nabla_{\theta_k} V$  computed by automatic differentiation through the composed map  $\theta_k \mapsto z_k \mapsto \pi_{S_{kj}}(z_k)$ .

**Lemma 3.2** (Per-Step Sufficient Decrease). *Fix  $W^t$  and let  $W^{t+1}$  follow the simultaneous update in Eq. (9). If  $V$  is  $L$ -smooth w.r.t. retraction on  $(\mathcal{W}, g)$  (Assumption 4.2) and  $0 < \eta_t \leq 1/L$ , then*

$$\begin{aligned} V(W^{t+1}) &\leq V(W^t) - \frac{\eta_t}{2} \|\text{grad } V(W^t)\|_g^2 \\ &= V(W^t) - \frac{\eta_t}{2} \sum_{i=1}^K \|\text{grad}_{g_i, \theta_i} V(W^t)\|_{g_i}^2. \end{aligned} \quad (10)$$

Telescoping this bound and using  $V \geq V_{\inf}$  (with  $\inf_t \eta_t > 0$ ) yields  $\lim_{t \rightarrow \infty} \|\text{grad } V(W^t)\|_g = 0$ , hence  $\|\text{grad}_{g_k, \theta_k} V(W^t)\|_{g_k} \rightarrow 0$  for every  $k$ ; every limit point of  $\{W^t\}$  is therefore a first-order Nash equilibrium. Equivalently, by Proposition 2.7, every client's payoff is simultaneously stationary ( $\text{grad}_{g_k, \theta_k} J_k \rightarrow 0$ ).

### 3.2. GeoEvo: Federated Geometric Evolution

Despite the first-order Nash stationarity guarantee of the reference solver, two structural barriers preclude its federated realization. (i) *Communication Infeasibility*. The pairwise term in  $V$  couples  $\theta_i$  and  $\theta_j$  across clients, yet each owns only its local parameter; exact computation of  $\text{grad}_{g_k, \theta_k} V$  thus demands per-step neighbor synchronization, contradicting the federated premise. (ii) *Multimodal Entrapment*. Composing non-convex losses  $\{F_k\}$  with pairwise distances over heterogeneous manifolds renders  $V$  deeply multimodal, so local block Riemannian descent (Eq. (9)) may remain confined to the basin of its initialization without nonlocal exploration. GeoEvo dissolves both via complementary server-client operations: at the server, per-round Fréchet prototypes compress  $\mathcal{O}(K^2)$  pairwise couplings into  $\mathcal{O}(K)$  per-client messages; at each client, population-level evolutionary search provides nonlocal exploration beyond local basins. Algorithm 1 formalizes the alternating procedure.

#### Server: Environment Field Construction.

Let  $z_k^t \triangleq \mathcal{T}_k(\theta_k^t)$  denote the structural representation of client  $k$  at round  $t$ , and let  $S^t \subseteq [K]$  be the set of clients participating in round  $t$ . The server collects  $\{z_k^t : k \in S^t\}$  and for each client  $k$ , forms the neighbor representation set  $\mathcal{E}_k^t \triangleq \{z_j^t : j \in \mathcal{N}_k^t\}$ ,  $\mathcal{N}_k^t \triangleq \mathcal{N}_k \cap S^t$ , where  $\mathcal{N}_k$  and the coupling sparsity  $\{\omega_{kj}, S_{kj}\}$  follow Assumption 2.6(iii).

**Fréchet Prototype.** To compress the per-pair projections of  $V$  in Eq. (3) while evaluating each neighbor only on its active overlap, we collapse the  $|\mathcal{N}_k^t|$  pairwise distance terms into a single Fréchet objective whose minimizer is the round- $t$  prototype:

$$Z_k^t \in \arg \min_{z \in \mathcal{Z}} \sum_{j \in \mathcal{N}_k^t} \omega_{kj} d_{\mathcal{Z}}^2(\pi_{S_{kj}}(z), \pi_{S_{kj}}(z_j^t)). \quad (11)$$

We summarize the neighborhood by its union support  $S_k^t \triangleq \bigcup_{j \in \mathcal{N}_k^t} S_{kj} \subseteq \mathcal{M}_k$  and aggregate weight  $\bar{\omega}_k^t \triangleq \sum_{j \in \mathcal{N}_k^t} \omega_{kj}$ , both used client-side below and well-defined there since  $S_k^t \subseteq \mathcal{M}_k$ . The minimizer  $Z_k^t$  is the weighted Fréchet mean of  $k$ 's neighbors on the identity-induced metric space  $(\mathcal{Z}, d_{\mathcal{Z}})$ , compressing all pairwise structural couplings  $\{(S_{kj}, \omega_{kj})\}_{j \in \mathcal{N}_k^t}$  into a single neighborhood centroid. Unlike class-centric prototypes,  $Z_k^t$  is a per-client *neighborhood centroid* driven by the game potential  $V$ .

**Single-Message Broadcast.** The server transmits the triple  $(Z_k^t, S_k^t, \bar{\omega}_k^t)$  as a single message to client  $k$ , achieving: (i) *Communication Compression*:  $\mathcal{O}(K^2)$  pairwise traffic is reduced to  $\mathcal{O}(K)$  aggregate-broadcast; (ii) *Privacy Preservation*: the server observes only structural representations  $\{z_j^t\}$ , never raw parameters  $\{\theta_j^t\}$ ; (iii) *Identity-Aware Aggregation*: the Fréchet mean on  $(\mathcal{Z}, d_{\mathcal{Z}})$  inherits the identity-induced geometry rather than enforcing a Euclidean mean.

#### Algorithm 1 GeoEvo: Federated Geometric Evolution

**Require:** Rounds  $T$ , local steps  $E$ , stepsizes  $(\eta, \eta_{\text{nes}})$ , samples  $m$ , coefficients  $(c_0, c_1, c_2)$ , damping  $\epsilon$

- 1: **Initialize:** for all  $k \in [K]$ :  $\theta_k \in \Theta_k$ ,  $u_k \leftarrow \mathbf{0}$ ,  $z_k^{\text{best}} \leftarrow \mathcal{T}_k(\theta_k)$
- 2: **for**  $t = 0, 1, \dots, T - 1$  **do**
- 3:   **Server:** sample  $S^t \subseteq [K]$ ; broadcast  $\{(Z_k^t, S_k^t, \bar{\omega}_k^t)\}_{k \in S^t}$  via Eq. (11)
- 4:   **for each**  $k \in S^t$  **in parallel do**
- 5:     **for**  $s = 1, \dots, E$  **do**
- 6:        $z_k \leftarrow \mathcal{T}_k(\theta_k)$
- 7:       **Descent:** compute  $\theta_k^{\text{base}}$  via Eq. (13)
- 8:       **NES:** compute  $\theta_k^{\text{NES}}$  via Eq. (15)
- 9:       **PSO:** compute  $\theta_k^{\text{PSO}}$  via Eq. (16)
- 10:       **Select:**  $\theta_k \leftarrow \theta_k^+$  via Eq. (17)
- 11:       **if**  $\phi_k(\mathcal{T}_k(\theta_k)) < \phi_k(z_k^{\text{best}})$  **then**  
          $z_k^{\text{best}} \leftarrow \mathcal{T}_k(\theta_k)$
- 12:     **end for**
- 13:     Upload  $z_k^{t+1} \leftarrow \mathcal{T}_k(\theta_k)$  to server
- 14:   **end for**
- 15: **end for**

#### Client: Manifold-Constrained Trajectory Search.

**Prototype-Based Surrogate.** The portion of  $V$  in Eq. (3) depending on  $\theta_k$  couples it to every neighbor through the pairwise sum  $\sum_{j \in \mathcal{N}_k^t} \omega_{kj} d_{\mathcal{Z}}^2(\pi_{S_{kj}}(z_k), \pi_{S_{kj}}(z_j^t))$ , which is prohibitive under federated communication: each per-pair projection  $\pi_{S_{kj}}$  demands a separate neighbor exchange. This many-to-many coupling collapses into a one-to-prototype surrogate:

$$\tilde{J}_k^t(\theta_k) \triangleq w_k F_k(\theta_k; \mathcal{I}_k) + \frac{\lambda}{2} \bar{\omega}_k^t d_{\mathcal{Z}}^2(\pi_{S_k^t}(z_k), \pi_{S_k^t}(Z_k^t)). \quad (12)$$

Since  $\tilde{J}_k^t$  depends only on the broadcast triple, each client minimizes it through purely local computation.

**Geometry-Aware Descent.** The Fisher-preconditioned retracted update of Eq. (9), originally defined for the exact potential  $V$ , transfers directly to the surrogate  $\tilde{J}_k^t$  by substituting  $\nabla_{\theta_k} \tilde{J}_k^t$  for  $\nabla_{\theta_k} V$ . This inheritance preserves the identity-induced block-sparsity of  $\tilde{\mathbf{F}}_k$  at no additional cost, ensuring the descent direction remains supported on  $\mathcal{M}_k$ -relevant coordinates. The resulting client-side Fisher-retracted step is given by:

$$\theta_k^{\text{base}} \leftarrow \text{Retr}_{\theta_k} \left( -\eta \tilde{\mathbf{F}}_k(\theta_k)^{-1} \nabla_{\theta_k} \tilde{J}_k^t(\theta_k) \right). \quad (13)$$

**Manifold-Lifted Evolutionary Search.**  $\tilde{J}_k^t$  inherits  $V$ 's multimodality, confining the descent in Eq. (13) to the basin of  $\theta_k^{\text{cur}}$ . This intrinsic limitation of first-order methods motivates evolutionary exploration in the block-Euclidean structure space  $\mathcal{Z}$ —a lower-dimensional, identity-aligned

domain whose global topology is more tractable than  $\Theta_k$ —with proposals lifted back to  $\Theta_k$  via the structure-matching operator

$$\text{Lift}_k(z; \theta_k) \in \arg \min_{\theta \in \Theta_k} d_{\mathcal{Z}}^2(\mathcal{T}_k(\theta), z). \quad (14)$$

The operator selects a closest structural match to  $z$ ; in practice, it is approximated by a few gradient steps warm-started from  $\theta_k$ .

**NES Proposal.** Let  $\phi_k(z) \triangleq \tilde{J}_k^t(\text{Lift}_k(z; \theta_k))$ . The Gaussian-smoothed gradient estimator is  $\hat{\nabla}_z \phi_{k, \sigma_z}(z_k) = \frac{1}{m\sigma_z} \sum_{i=1}^m \phi_k(z_k + \sigma_z \varepsilon_i) \varepsilon_i$  with  $\varepsilon_i \sim \mathcal{N}(0, \mathbf{I})$ , yielding the NES proposal

$$\theta_k^{\text{NES}} \leftarrow \text{Lift}_k(z_k - \eta_{\text{nes}} \hat{\nabla}_z \phi_{k, \sigma_z}(z_k); \theta_k). \quad (15)$$

**PSO Proposal.** Client  $k$  maintains a velocity  $u_k \in \mathcal{Z}$  and personal best  $z_k^{\text{best}}$ ; setting the broadcast prototype  $Z_k^t$  as the swarm’s global attractor embeds cross-client structural coordination directly into the swarm dynamics.

$$\begin{aligned} u_k &\leftarrow c_0 u_k + c_1 r_1 (z_k^{\text{best}} - z_k) + c_2 r_2 (Z_k^t - z_k), \\ \theta_k^{\text{PSO}} &\leftarrow \text{Lift}_k(z_k + u_k; \theta_k). \end{aligned} \quad (16)$$

**Monotone Selection.** At each local step, GeoEvo evaluates the descent candidate  $\theta_k^{\text{base}}$  alongside the evolutionary proposals  $\{\theta_k^{\text{NES}}, \theta_k^{\text{PSO}}\}$ , retaining the surrogate-minimizer:

$$\theta_k^+ \leftarrow \arg \min_{\theta \in \{\theta_k^{\text{cur}}, \theta_k^{\text{base}}, \theta_k^{\text{NES}}, \theta_k^{\text{PSO}}\}} \tilde{J}_k^t(\theta), \quad (17)$$

and updates the personal best  $z_k^{\text{best}} \leftarrow \mathcal{T}_k(\theta_k^+)$  when this strictly decreases  $\phi_k$ . Including the current iterate  $\theta_k^{\text{cur}}$  in the candidate pool guarantees that  $\tilde{J}_k^t(\theta_k^+) \leq \tilde{J}_k^t(\theta_k^{\text{cur}})$ . This condition transfers the per-step decrease of Lemma 3.2 from  $V$  to the surrogate, remaining robust to any inexactness in the structural  $\text{Lift}_k$  operator since selection always evaluates the exact surrogate objective  $\tilde{J}_k^t$  across all candidates.

Full proofs and technical details are deferred to Appendix C.

## 4. Theoretical Analysis

Our theoretical analysis bridges the idealized centralized reference solver and its communication-feasible federated realization. For the exact potential  $V$  in Eq. (3), local minimizers are local Nash equilibria (Proposition 4.4) and retracted block descent on  $(\mathcal{W}, g)$  yields sufficient decrease and first-order stationarity (Theorem 4.5). For its federated realization, the server prototype is Fréchet-optimal for the exact per-pair coupling (Proposition 4.6); under partial participation and bounded inter-round surrogate drift, the GeoEvo client update attains an  $\mathcal{O}(1/\sqrt{T})$  surrogate-stationarity rate despite its monotone selection (Theorem 4.10); and NES/PSO proposals preserve monotone surrogate descent within each round (Proposition 4.12).

### 4.1. Preliminaries and Assumptions

We endow the product space  $\mathcal{W} = \prod_{k=1}^K \Theta_k$  with the product metric  $g \triangleq \oplus_{k=1}^K g_k$ , and write  $\text{grad } V(W)$  for the Riemannian gradient on  $(\mathcal{W}, g)$  and  $\text{grad}_{g_k, \theta_k} V(W)$  for its  $k$ -th block. All retractions are denoted by  $\text{Retr}$ .

**Assumption 4.1** (Geometry and Regularity). For each client  $k$ ,  $\Theta_k$  admits a second-order retraction  $\text{Retr}$ , and  $F_k(\cdot; \mathcal{I}_k)$  and  $\mathcal{T}_k$  are twice continuously differentiable. For each edge  $\{k, j\}$  with  $S_{kj} \neq \emptyset$ , the map  $\theta_k \mapsto d_{\mathcal{Z}}^2(\pi_{S_{kj}}(\mathcal{T}_k(\theta_k)), \pi_{S_{kj}}(\mathcal{T}_j(\theta_j)))$  is continuously differentiable in  $\theta_k$  for any fixed  $\theta_j$ .

**Assumption 4.2** (Smoothness on Manifolds). There exists  $L > 0$  such that the potential  $V$  on  $(\mathcal{W}, g)$  and every round- $t$  client surrogate  $\tilde{J}_k^t$  on  $(\Theta_k, g_k)$  are  $L$ -smooth w.r.t.  $\text{Retr}$ . For any  $W \in \mathcal{W}$  and  $\xi \in T_W \mathcal{W}$ ,

$$V(\text{Retr}_W(\xi)) \leq V(W) + \langle \text{grad } V(W), \xi \rangle_g + \frac{L}{2} \|\xi\|_g^2,$$

and analogously

$$\tilde{J}_k^t(\text{Retr}_{\theta_k}(\xi_k)) \leq \tilde{J}_k^t(\theta_k) + \langle \text{grad}_{g_k} \tilde{J}_k^t(\theta_k), \xi_k \rangle_{g_k} + \frac{L}{2} \|\xi_k\|_{g_k}^2$$

for  $\xi_k \in T_{\theta_k} \Theta_k$ .

**Assumption 4.3** (Lower Boundedness and Compactness).

- (i)  $V$  is bounded below on  $\mathcal{W}$ , i.e.,  $V(W) \geq V_{\text{inf}} > -\infty$ .
- (ii) (Optional, for limit-point statements) the sublevel set  $\mathcal{L}(V(W^0)) = \{W \in \mathcal{W} : V(W) \leq V(W^0)\}$  is compact.

### 4.2. Exact Potential and First-Order Nash Equilibria

**Proposition 4.4** (Exact Potential Game (Restated)). Under Assumption 2.6(iii) and the symmetric edge convention ( $\omega_{kj} = \omega_{jk}$  and the same  $S_{kj}$  and  $\pi_{S_{kj}}$  are used on edge  $\{k, j\}$ ),  $V$  in Eq. (3) is an exact potential for the game in Eq. (2), i.e., Eq. (4) holds for any unilateral deviation. Consequently, any local minimizer of  $V$  on  $\mathcal{W}$  is a local Nash equilibrium.

*Proof Sketch.* Fix client  $k$  and  $W_{-k}$ , and consider a unilateral deviation  $\theta_k \rightarrow \theta'_k$ . The terms of  $V(W)$  affected by  $\theta_k$  are the local term  $w_k F_k(\theta_k; \mathcal{I}_k)$  and the pairwise terms on edges  $\{k, j\}$ . Since  $V$  counts each undirected edge once, the induced change in  $V$  equals

$$\begin{aligned} &w_k (F_k(\theta'_k; \mathcal{I}_k) - F_k(\theta_k; \mathcal{I}_k)) \\ &+ \frac{\lambda}{2} \sum_{j \in \mathcal{N}_k} \omega_{kj} \left( d_{\mathcal{Z}}^2(\pi_{S_{kj}}(\mathcal{T}_k(\theta'_k)), \pi_{S_{kj}}(\mathcal{T}_j(\theta_j))) \right. \\ &\quad \left. - d_{\mathcal{Z}}^2(\pi_{S_{kj}}(\mathcal{T}_k(\theta_k)), \pi_{S_{kj}}(\mathcal{T}_j(\theta_j))) \right), \end{aligned} \quad (18)$$

which is exactly  $J_k(\theta'_k; W_{-k}) - J_k(\theta_k; W_{-k})$  by the symmetry  $\omega_{kj} = \omega_{jk}$  and the use of the same overlap operator  $\pi_{S_{kj}}$  on each edge.  $\square$

**First-Order Nash Equilibrium (FONE).** If  $\text{grad } V(W^*) = 0$  (equivalently,  $\text{grad}_{g_k, \theta_k} V(W^*) = 0$  for every  $k$ ), we call  $W^*$  a *first-order Nash equilibrium* (FONE), i.e., each player satisfies a first-order stationarity condition under its feasible geometry. FONE is necessary but not sufficient for local Nash optimality.

### 4.3. Retracted Block Descent and Stationarity

**Theorem 4.5** (Stationarity Convergence (Simultaneous Blocks)). *Under Assumptions 4.2–4.3, for the iterates  $\{W^t\}$  of the block descent in Eq. (9) with stepsizes  $0 < \eta_t \leq 1/L$ , the sequence  $\{V(W^t)\}$  is non-increasing and  $\sum_{t=0}^{\infty} \eta_t \|\text{grad } V(W^t)\|_g^2 < \infty$ . In particular, if  $\inf_t \eta_t > 0$ , then  $\lim_{t \rightarrow \infty} \|\text{grad } V(W^t)\|_g = 0$ ; under Assumption 4.3(ii) and the continuity of  $\text{grad } V$  (ensured by the  $L$ -smoothness in Assumption 4.2), every limit point of  $\{W^t\}$  is a FONE.*

**From the Reference Solver to GeoEvo.** Theorem 4.5 characterizes the *idealized centralized* reference solver, which evaluates the exact gradient  $\text{grad } V$  and therefore requires per-step neighbor synchronization. GeoEvo replaces this exact coupling by stale broadcast prototypes under partial participation  $S^t$ ; deterministic monotone decrease of  $V$  is therefore not guaranteed in general for coupled non-convex problems. The federated guarantees relevant to GeoEvo are the per-round surrogate stationarity of Theorem 4.10 and the monotone selection of Proposition 4.12.

### 4.4. Optimal Surrogate Approximation

Fix round  $t$  and recall  $\mathcal{N}_k^t = \mathcal{N}_k \cap S^t$ ,  $S_k^t = \bigcup_{j \in \mathcal{N}_k^t} S_{kj} \subseteq \mathcal{M}_k$ , and  $\bar{\omega}_k^t = \sum_{j \in \mathcal{N}_k^t} \omega_{kj}$ . We assume  $\mathcal{Z}$  is a product space and  $d_{\mathcal{Z}}$  acts coordinatewise under masking, so that  $\pi_S(\pi_{S'}(\cdot)) = \pi_S(\cdot)$  whenever  $S \subseteq S'$ . The per-pair coupling of  $V$  attributable to client  $k$  is  $\Phi_k^t(z) \triangleq \frac{\lambda}{2} \sum_{j \in \mathcal{N}_k^t} \omega_{kj} d_{\mathcal{Z}}^2(\pi_{S_{kj}}(z), \pi_{S_{kj}}(z_j^t))$ .

**Proposition 4.6** (Variational Optimality of Prototypes). *The server-computed prototype  $Z_k^t$  in Eq. (11) satisfies  $\Phi_k^t(Z_k^t) \leq \Phi_k^t(z)$  for all  $z \in \mathcal{Z}$ , i.e.,  $Z_k^t$  minimizes the exact per-pair coupling  $\Phi_k^t$ . By coordinatewise separation,  $Z_k^t$  is, on each modality  $m \in S_k^t$ , the weighted Fréchet mean over the neighbors that possess  $m$ .*

To reduce communication, GeoEvo replaces  $\Phi_k^t$  with the single-prototype surrogate that collapses the  $|\mathcal{N}_k^t|$  per-pair terms into a single prototype  $Z_k^t$ :

$$\tilde{\Phi}_k^t(z) \triangleq \frac{\lambda}{2} \bar{\omega}_k^t d_{\mathcal{Z}}^2(\pi_{S_k^t}(z), \pi_{S_k^t}(Z_k^t)). \quad (19)$$

**Assumption 4.7** (Surrogate Approximation Error). *There exists  $\delta_k^t \geq 0$  such that for all  $\theta_k \in \Theta_k$ ,  $|\Phi_k^t(\mathcal{T}_k(\theta_k)) - \tilde{\Phi}_k^t(\mathcal{T}_k(\theta_k))| \leq \delta_k^t$ . This bound captures*

two effects: the prototype-compression error (neighbor dispersion on  $S_k^t$ ), and the residual from collapsing the per-pair projections  $\{\pi_{S_{kj}}\}_j$  and weights  $\{\omega_{kj}\}_j$  onto the single union mask  $\pi_{S_k^t}$  and aggregate weight  $\bar{\omega}_k^t$ .

### 4.5. Stochastic Convergence to Surrogate Stationarity

For practical federated settings, we use stochastic gradient estimators to locally optimize  $\tilde{J}_k^t$ . Define  $\tilde{J}^t \triangleq |S^t|^{-1} \sum_{k \in S^t} \tilde{J}_k^t(\theta_k^t)$ , and let  $\tilde{J}_{\inf} > -\infty$  be a uniform lower bound on round-input and round-output averages.

**Assumption 4.8** (Stochastic Gradients). *Each participating client employs an unbiased stochastic gradient estimator  $\widehat{\text{grad}}_{g_k, \theta_k} \tilde{J}_k^t$  with bounded variance:*

$$\mathbb{E}[\widehat{\text{grad}}_{g_k, \theta_k} \tilde{J}_k^t \mid \theta_k, Z_k^t, S_k^t, \bar{\omega}_k^t] = \text{grad}_{g_k, \theta_k} \tilde{J}_k^t, \quad (20)$$

$$\mathbb{E}[\|\widehat{\text{grad}}_{g_k, \theta_k} \tilde{J}_k^t - \text{grad}_{g_k, \theta_k} \tilde{J}_k^t\|_{g_k}^2 \mid \theta_k, Z_k^t, S_k^t, \bar{\omega}_k^t] \leq \sigma^2. \quad (21)$$

**Assumption 4.9** (Bounded Inter-Round Surrogate Drift). *Let  $\xi_t \triangleq \tilde{J}^{t+1} - \frac{1}{|S^t|} \sum_{k \in S^t} \tilde{J}_k^t(\theta_k^{t+1})$  and  $[a]_+ \triangleq \max\{a, 0\}$ . The cumulative adverse inter-round change is bounded in expectation:  $\sum_{t=0}^{T-2} \mathbb{E}[[\xi_t]_+] \leq \Delta_T$ .*

**Theorem 4.10** (Non-Convex Convergence Rate (Surrogate Stationarity)). *Under Assumptions 4.2, 4.8, and 4.9, consider the iterates generated by the GeoEvo client update (17) with stepsize  $\eta \leq 1/L$ , whose candidate pool contains a base candidate computed from an unbiased stochastic gradient estimator. An absolute constant  $C > 0$  then satisfies*

$$\min_{t < T} \mathbb{E} \left[ \frac{1}{|S^t|} \sum_{k \in S^t} \|\text{grad}_{g_k, \theta_k} \tilde{J}_k^t\|_{g_k}^2 \right] \leq \frac{C(\mathbb{E}[\tilde{J}^0] - \tilde{J}_{\inf} + \Delta_T)}{\eta T} + CL\eta\sigma^2. \quad (22)$$

*Under summable inter-round surrogate drift, Assumption 4.9 gives  $\Delta_T = \mathcal{O}(1)$ . Setting  $\eta = \Theta(1/\sqrt{T})$  yields an  $\mathcal{O}(1/\sqrt{T})$  rate; if  $\Delta_T = o(\sqrt{T})$ , bound is  $o(1)$ , whereas  $\Delta_T = \Theta(\sqrt{T})$  leaves an  $\mathcal{O}(1)$  stationarity neighborhood.*

**Proof Sketch.** At the first local update, the candidate pool in Eq. (17) contains the base candidate computed from an unbiased stochastic gradient estimator. Hence the first accepted iterate inherits the Riemannian-SGD descent bound. Since each subsequent selection retains the current iterate, the final round output after  $E$  local steps is no worse than the first accepted iterate. Telescoping over  $T$  rounds with inter-round drift absorbed by  $\Delta_T$  gives Eq. (22).  $\square$

**Remark 4.11** (Expected Dissipation of  $V$ ). *Under the federated-lift and gradient-consistency conditions of Proposition D.10, the surrogate guarantee lifts to a conditional expected-dissipation inequality for exact potential  $V$ . The resulting first-order-stationarity neighborhood is controlled by the surrogate-approximation and partial-participation gaps, together with the assumed cumulative lift residual.*

Table 1. Performance comparison on CREMA-D across varying missing rates in terms of accuracy (%).

| Method                                   | 30%                     | 50%                     | 70%                     |
|------------------------------------------|-------------------------|-------------------------|-------------------------|
| <b>Baselines</b>                         |                         |                         |                         |
| Centralized                              | 54.10 $\pm$ 0.80        | 52.30 $\pm$ 0.95        | 50.95 $\pm$ 1.40        |
| IID                                      | 52.07 $\pm$ 2.06        | 51.29 $\pm$ 1.99        | 49.60 $\pm$ 2.02        |
| <b>FL (Global &amp; Local Trade-off)</b> |                         |                         |                         |
| FedProx (Li et al., 2020)                | 47.30 $\pm$ 0.94        | 46.86 $\pm$ 1.38        | 48.59 $\pm$ 2.07        |
| SCAFFOLD (Karimireddy et al., 2020)      | 50.51 $\pm$ 3.00        | 48.67 $\pm$ 2.02        | 47.14 $\pm$ 2.67        |
| HetPFL (Ye & Tang, 2025)                 | 48.47 $\pm$ 1.52        | 47.38 $\pm$ 1.05        | 46.50 $\pm$ 3.71        |
| FedMoSWA (Liu et al., 2025)              | 50.68 $\pm$ 1.21        | 49.61 $\pm$ 1.37        | 48.56 $\pm$ 3.65        |
| <b>PFL (Euclidean-Constrained)</b>       |                         |                         |                         |
| Ditto (Li et al., 2021)                  | 47.57 $\pm$ 0.94        | 46.94 $\pm$ 0.99        | 43.14 $\pm$ 2.46        |
| HB-PFL (Thapa & Li, 2025)                | 51.18 $\pm$ 0.88        | 48.40 $\pm$ 1.22        | 46.95 $\pm$ 2.21        |
| pFedMoAP (Luo et al., 2025)              | 51.91 $\pm$ 1.54        | 50.37 $\pm$ 1.05        | 47.50 $\pm$ 2.90        |
| FedMGP (Bo et al., 2026)                 | 51.60 $\pm$ 0.97        | 49.55 $\pm$ 2.15        | 47.10 $\pm$ 4.48        |
| <b>MMFL (Compensation-oriented)</b>      |                         |                         |                         |
| FedMultimodal (Feng et al., 2023)        | 48.88 $\pm$ 0.68        | 45.30 $\pm$ 1.88        | 44.82 $\pm$ 2.15        |
| $M^3$ Fed (Li et al., 2024)              | 50.64 $\pm$ 0.88        | 49.85 $\pm$ 1.58        | 47.42 $\pm$ 2.30        |
| FedMEMA (Dai et al., 2024)               | 51.15 $\pm$ 0.90        | 50.10 $\pm$ 2.25        | 49.35 $\pm$ 2.32        |
| FED-PRIME (Phung et al., 2025)           | 50.95 $\pm$ 1.38        | 49.07 $\pm$ 1.40        | 48.08 $\pm$ 2.54        |
| PEPSY (Nguyen et al., 2026)              | 52.02 $\pm$ 0.85        | 50.85 $\pm$ 1.44        | 48.90 $\pm$ 3.05        |
| <b>PMFL (Proposed Identity-Aware)</b>    |                         |                         |                         |
| PMFL (Vanilla)                           | 50.31 $\pm$ 1.26        | 49.69 $\pm$ 1.28        | 47.54 $\pm$ 2.46        |
| Oracle (Exact)                           | <b>56.40</b> $\pm$ 0.85 | 56.85 $\pm$ 1.10        | 54.10 $\pm$ 1.95        |
| GeoEvo (Prototype)                       | 55.82 $\pm$ 0.92        | 53.90 $\pm$ 1.40        | 52.05 $\pm$ 3.50        |
| GeoEvo (Ours)                            | 56.06 $\pm$ 1.03        | <b>57.56</b> $\pm$ 1.63 | <b>54.67</b> $\pm$ 2.14 |

#### 4.6. Monotone Surrogate Descent and Exploration

**Proposition 4.12** (Evolutionary Proposals Preserve Monotonicity). *Fix round  $t$  and fields  $\{Z_k^t\}$ . Let  $\theta_k^t$  be any proposal (gradient-based or evolutionary). If the client uses the selection rule in Eq. (17), which retains the current iterate  $\theta_k^t$  (denoted  $\theta_k^{\text{cur}}$  therein) in the candidate pool, then  $\tilde{J}_k^t(\theta_k^{t+1}) \leq \tilde{J}_k^t(\theta_k^t)$ . This holds unconditionally: the filter evaluates the exact parameter-space surrogate  $\tilde{J}_k^t(\cdot)$ , so it is immune to any optimization inexactness in the structural Lift<sub>k</sub> operator, which at worst yields a weaker candidate. Under Assumption 4.7, the corresponding exact per-pair coupling satisfies  $\Phi_k^t(\mathcal{T}_k(\theta_k^{t+1})) \leq \Phi_k^t(\mathcal{T}_k(\theta_k^t)) + 2\delta_k^t$ .*

**Remark 4.13** (Saddle Avoidance in Practice). The stochastic exploratory nature of NES/PSO proposals, combined with the monotone selection rule, facilitates identifying descent directions in regions of negative curvature, empirically mitigating stagnation at saddle points.

Full proofs and supporting lemmas are in Appendix D.

## 5. Experiments

**Experimental Setup.** We evaluate GeoEvo on three multimodal benchmarks: CREMA-D (Cao et al., 2014) (audio-visual emotion), CrisisMMD (Alam et al., 2018) (image-text crisis), and the large-scale UPMC Food-101 (Wang et al., 2015) (image-text food). To simulate federated het-

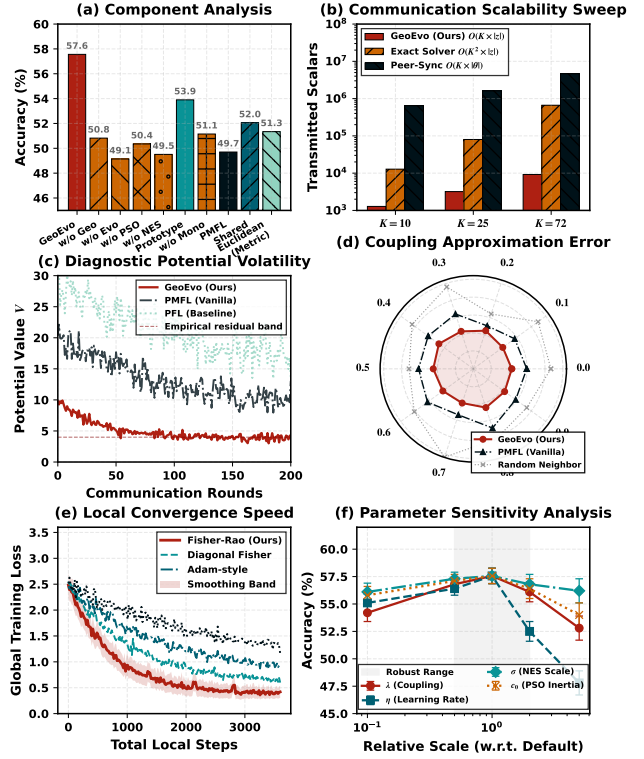


Figure 1. Ablation of GeoEvo on CREMA-D at 50% missing rate.

erogeneity, we partition CREMA-D by speaker identity, while the other two datasets are split using a Dirichlet prior  $\text{Dir}(\alpha)$ . Modality incompleteness is introduced by two patterns: proportional (dataset-native) and binomial (client-wise Bernoulli). Following (Feng et al., 2023; Phung et al., 2025), we employ heterogeneous backbones (e.g., MobileNetV2, MobileBERT) for small-scale tasks and ViLT for UPMC Food-101. Each client reserves a portion of its local data as a private test set to evaluate the model’s alignment with its personalized distribution. We compare against 15 representative baselines spanning centralized, FL, PFL, and MMFL paradigms, as well as our proposed PMFL variants.

**Overall Performance Superiority.** GeoEvo achieves the strongest overall performance across three benchmarks, with gains becoming most pronounced under severe modality incompleteness. On CREMA-D (Table 1), it achieves 54.67% under a 70% missing rate, numerically surpassing the Oracle in mean accuracy as incompleteness deepens. This advantage extends to CrisisMMD (Figure 2) and UPMC Food-101 (Figure 3), where GeoEvo remains top-tier across heterogeneity levels and degrades far more gracefully than FL, PFL, and MMFL methods as missing rates rise. This superiority stems from a mechanistic shift: modality incompleteness is an intrinsic structural identity to be respected, not a deficiency to be repaired, thereby guiding geometry-aware adaptation along each client’s identifiable coordinates.

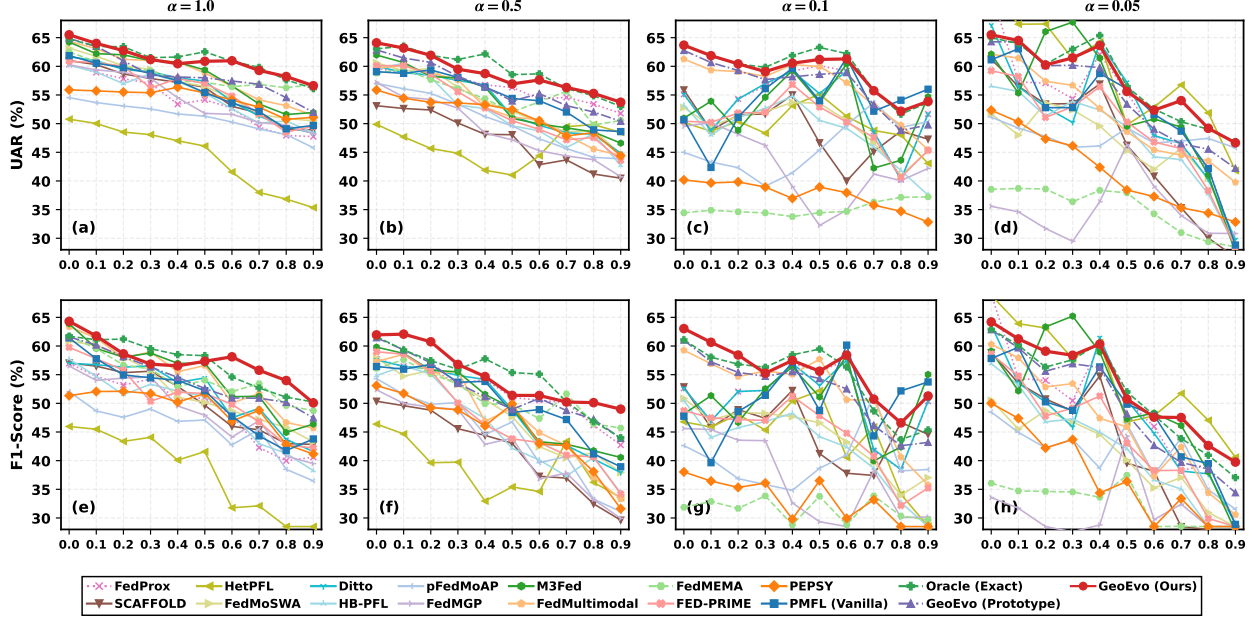


Figure 2. Comparative performance on CrisisMMD under binomial modality simulation across varying heterogeneity levels.

By casting PMFL as an identity-aware potential game, GeoEvo replaces semantically ill-posed shared-coordinate averaging with adaptation on client-specific Riemannian submanifolds equipped with Fisher metrics that inherit each client’s modality block-sparsity to nullify unobserved-dimension updates and avoid the imputation noise that limits compensation-oriented MMFL. This is why the baselines forfeit ground: FL fuses dimensionally incompatible optima into a single model, Euclidean PFL injects unidentifiable directions that drive negative transfer, and MMFL mistakes a permanent structural void for a transient shift, incurring an irreducible optimality gap. Conversely, GeoEvo coordinates via server-broadcast Fréchet prototypes that preserve heterogeneous equilibria, with a monotone selection rule making nonlocal exploration safe by construction.

**Ablation and Diagnostic Analysis.** Figure 1(a) isolates each component: removing evolutionary search (*w/o Evo*) or geometric coupling (*w/o Geo*) causes drops of 8.5 and 6.8 pp from the full 57.6%. Ablating either evolutionary operator, the monotone rule, or the Fisher metric reduces accuracy, confirming the complementary contribution of every design component. These benefits incur no additional pairwise communication: Figure 1(b) shows GeoEvo’s traffic scaling linearly as  $\mathcal{O}(K|z|)$ , far below the exact solver’s quadratic  $\mathcal{O}(K^2|z|)$  coupling. The potential trajectory (Figure 1(c)) shows GeoEvo contracting  $V$  into an empirical residual band and fluctuating within it, consistent with the conditional dissipation behavior in Proposition D.10 under stale prototypes and partial participation. The monotone rule (Proposition 4.12) bounds exploration volatility, whereas

the uncoupled vanilla PMFL and PFL baselines exhibit persistent, non-vanishing fluctuations. Across missing-rate regimes, Figure 1(d) keeps GeoEvo’s coupling error lowest, validating the variational optimality of the Fréchet prototype (Proposition 4.6) over vanilla and random-neighbor surrogates; Figure 1(e) shows the Fisher–Rao update converging faster than diagonal-Fisher and Adam-style preconditioners; and Figure 1(f) confirms robustness across hyperparameters, with sensitivity concentrated on the learning rate, consistent with the stepsize condition  $\eta \leq 1/L$ .

Full configurations and results are deferred to Appendix E.

## 6. Conclusion

This paper reconceptualizes PMFL by treating modality incompleteness as an intrinsic structural identity, formalizing cross-client collaboration as an Identity-Aware Potential Game realized through GeoEvo, which integrates curvature-adaptive Fisher descent with manifold-lifted evolutionary search under Fréchet prototypes. GeoEvo attains monotone surrogate descent and an  $\mathcal{O}(1/\sqrt{T})$  stationarity rate under bounded inter-round drift with  $\mathcal{O}(K)$  messages per communication, empirically improving personalization and robustness across missing-modality regimes. GeoEvo’s single-prototype compression trades a minor surrogate-approximation residual inherent to distributed non-convex landscapes for linear message complexity. Tightening this controlled residual via variance-reduced prototypes and extending the potential-game framework to asynchronous client populations represent promising directions.

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## Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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## A. Related Works

### A.1. Federated Learning

Federated learning (FL) enables collaborative training without centralizing raw data, but client heterogeneity often causes biased optimization and unstable convergence. Classic FedAvg (McMahan et al., 2017) and its heterogeneity-robust variants address client drift via regularization (FedProx (Li et al., 2020)) or control variates (SCAFFOLD (Karimireddy et al., 2020)). However, these approaches assume homogeneous input spaces and model architectures, which becomes restrictive in multimodal settings where clients suffer permanent structural missingness. Our work builds on FL by explicitly modeling each client’s structural identity and enabling geometry-consistent coupling across heterogeneous submanifolds.

### A.2. Personalized Federated Learning

Personalized Federated Learning (PFL) addresses statistical heterogeneity by learning client-adaptive models rather than enforcing a single global solution. Existing PFL methods are commonly grouped into: (i) **Regularization-based** approaches that constrain local drift from a reference model (e.g., Ditto (Li et al., 2021)); (ii) **Model decoupling or partial sharing** methods that separate globally shared representations from client-specific components (e.g., FedPer (Arivazhagan et al., 2019), FedRep (Collins et al., 2021)); (iii) **Meta-learning** frameworks that learn an initialization enabling fast client adaptation (e.g., Per-FedAvg (Fallah et al., 2020)); and (iv) **Prototype or distillation-based** techniques that exchange lightweight knowledge to improve personalization without sharing full parameters (e.g., FedProto (Tan et al., 2022b)). While effective in unimodal settings, these paradigms typically assume a shared parameter geometry and a comparable notion of distance across clients. In contrast, our setting treats permanent modality absence as an intrinsic structural identity, which induces heterogeneous feasible geometries and makes naive global consensus or modality-agnostic aggregation prone to representational distortion.

### A.3. Multimodal Federated Learning

Multimodal Federated Learning (MMFL) extends FL to scenarios where each client observes multiple modalities, often with severe modality imbalance and missingness. Most existing solutions treat missing modalities as a deficiency to be compensated toward a full-modality template, which can be broadly grouped into three paradigms: (i) **Imputation-based** methods that reconstruct missing modalities using external resources (e.g., public datasets (Yu et al., 2023), prototype libraries (Saha et al., 2025), reconstruction models (Wang et al., 2024), or pre-trained foundation models (Qiao et al., 2025)); (ii) **Disentanglement-based** methods that partition architectures into modality-specific branches and coordinate them via auxiliary alignment or distillation (Peng et al., 2024; Ouyang et al., 2023; Sun et al., 2024; Chen & Zhang, 2024); and (iii) **Aggregation-based** methods that design structured message passing or block-wise aggregation among multimodal clients (Chen & Zhang, 2022; Qi & Li, 2024; Borazjani et al., 2024). While effective when missingness is transient or recoverable, these compensation-centric objectives can conflate permanent client-specific absence with distribution shift, encouraging artificial correspondence and amplifying aggregation bias. In contrast, we formalize modality configurations as identities and couple clients only through shared modality supports in a common structure space, avoiding alignment on unsupported dimensions—thereby preserving each client’s intrinsic optimization geometry. This stands in sharp contrast to prior MMFL methods that implicitly enforce a shared Euclidean template.

### A.4. Game-Theoretic and Evolutionary Perspectives in Federated Optimization

Game-theoretic formulations model strategic interactions among heterogeneous agents; potential games are particularly attractive because equilibria can be characterized via a single scalar potential function (Monderer & Shapley, 1996). This perspective aligns naturally with personalization: clients may converge to distinct solutions while retaining structured coupling. In parallel, evolutionary computation provides gradient-free or hybrid exploration mechanisms for non-convex optimization, including Natural Evolution Strategies (NES) (Wierstra et al., 2014) and swarm-based methods such as Particle Swarm Optimization (PSO) (Kennedy & Eberhart, 1995). These tools have been adopted in distributed learning and FL to improve robustness, exploration–exploitation balance, and resilience to noisy objectives (e.g., population-based training (Jaderberg et al., 2017)). Our method connects these threads by (i) casting identity-aware Personalized Multimodal Federated Learning (PMFL) as an exact potential game under symmetric coupling on shared supports, (ii) designing a geometry-aware evolutionary optimizer that respects client-specific information geometry while enforcing monotone decrease of a local surrogate objective via an selection rule, and (iii) compressing the  $\mathcal{O}(K^2)$  pairwise coupling to  $\mathcal{O}(K)$  communication via server-broadcast Fréchet prototypes.

## B. Appendix for the Problem Formulation

**Notation.**  $\mathcal{M} = \{m_1, \dots, m_M\}$  is the modality universe. Client  $k$  has a fixed modality set  $\mathcal{M}_k \subseteq \mathcal{M}$  and identity  $\mathcal{I}_k = (\mathcal{D}_k, \mathcal{M}_k)$  (Definition 2.4). Its feasible geometry is the manifold constraint set  $\Theta_k$  with intrinsic metric  $g_k$ . The joint strategy profile is  $W = (\theta_1, \dots, \theta_K) \in \mathcal{W} \triangleq \prod_{k=1}^K \Theta_k$ , and  $W_{-k}$  omits the  $k$ -th block. For a pair  $(k, j)$  the shared support is  $S_{kj} \triangleq \mathcal{M}_k \cap \mathcal{M}_j$  and  $\mathcal{N}_k \triangleq \{j \in [K] \setminus \{k\} : S_{kj} \neq \emptyset\}$ . The structure-extraction map is  $\mathcal{T}_k : \Theta_k \rightarrow \mathcal{Z}$  with  $z_k \triangleq \mathcal{T}_k(\theta_k)$ , the restriction map is  $\pi_S : \mathcal{Z} \rightarrow \mathcal{Z}_S$ , and  $d_{\mathcal{Z}}$  denotes the metric on each  $\mathcal{Z}_S$ . Coupling weights satisfy  $\omega_{kj} = \omega_{jk} \geq 0$  with  $\omega_{kj} > 0 \iff S_{kj} \neq \emptyset$  (Assumption 2.6(iii)). We write  $\langle \cdot, \cdot \rangle_{g_k}, \|\cdot\|_{g_k}$  for the metric on  $T_{\theta_k} \Theta_k$  and  $\text{grad}_{g_k, \theta_k}$  for the Riemannian gradient in the  $k$ -th block. Throughout, “ $Y \not\perp\!\!\!\perp X \mid Z$ ” denotes conditional dependence.

### B.1. Static vs. Dynamic Absence: Justifying the Structural Identity

Definition 2.2 promotes the modality set  $\mathcal{M}_k$  from a statistical accident of  $\mathcal{D}_k$  to a permanent attribute of the client. The following regime distinction makes precise the constraint that licenses this promotion.

**Definition B.1** (Modality-Absence Regimes). Let  $\mathcal{M}_k^{(t)}$  be the modalities available to client  $k$  at round  $t$ .

- **Dynamic Absence:**  $\exists t \neq t'$  with  $\mathcal{M}_k^{(t)} \neq \mathcal{M}_k^{(t')}$  (transient occlusion, sensor dropout).
- **Static Absence:**  $\mathcal{M}_k^{(t)} = \mathcal{M}_k$  for all  $t \geq 0$ , reflecting time-invariant hardware, policy, or cost constraints.

**Proposition B.2** (Mis-Specification of Full-Modality Consensus under Static Absence). *Assume static absence and a modality-factorized admissible model class in which blocks outside  $\mathcal{M}_k$  are inactive for client  $k$ . Any objective term whose dependence on  $\theta_k$  is only through an inactive modality block is constant over the admissible local model class and therefore contributes zero local gradient.*

*Proof.* Under static absence, client  $k$ ’s prediction depends only on  $\mathbf{x}_{\mathcal{M}_k}$ , and every branch  $m \notin \mathcal{M}_k$  is inactive. Hence the derivative of the prediction, and consequently of any local objective term, with respect to the corresponding parameter block is identically zero. A full-modality consensus term requiring information from such a branch is therefore unsupported by the client’s admissible observation model.  $\square$

This is precisely the failure of the vanilla extension Eq. (1): it treats  $\mathcal{M} \setminus \mathcal{M}_k$  as missing data of  $\mathcal{D}_k$  rather than as a structural invariant, and couples clients through a Euclidean consensus that ignores  $\mathcal{M}_k$ . The remedy adopted in the main text is to couple clients only through the shared supports  $S_{kj}$  and to keep  $\mathcal{M}_k$  invariant.

### B.2. The Irreducible Optimality Gap

We formalize the inequality  $F_k^*(\mathcal{M}_k) \geq F_k^*(\mathcal{M})$  asserted in the “Irreducible Optimality Gap” paragraph. For a modality subset  $\mathcal{M}' \subseteq \mathcal{M}$  write  $F_k(\theta; \mathcal{M}') \triangleq \mathbb{E}_{\mathcal{D}_k | \mathcal{M}'}[\ell(f_\theta(\mathbf{x}_{\mathcal{M}'}), Y)]$  and  $F_k^*(\mathcal{M}') \triangleq \inf_{\theta \in \Theta} F_k(\theta; \mathcal{M}')$ .

**Lemma B.3** (Monotonicity of Bayes Risk in the Observation  $\sigma$ -Field). *Let  $\ell$  be a strictly proper loss whose Bayes act given a  $\sigma$ -field  $\mathcal{G}$  is the  $\mathcal{G}$ -conditional functional of  $Y$  (e.g., squared loss with  $\mathbb{E}[Y \mid \mathcal{G}]$ , or log loss with  $P(Y \mid \mathcal{G})$ ). If the hypothesis class  $\{f_\theta\}_{\theta \in \Theta}$  is expressive enough to realize these conditional functionals, then for  $\mathcal{M}_k \subseteq \mathcal{M}$ ,*

$$F_k^*(\mathcal{M}_k) \geq F_k^*(\mathcal{M}), \quad (23)$$

*with strict inequality exactly when the finer observation  $\sigma$ -field changes the Bayes act on a set of positive probability.*

*Proof.* Let  $\mathcal{G}_k \triangleq \sigma(X_{\mathcal{M}_k}) \subseteq \sigma(X_{\mathcal{M}}) \triangleq \mathcal{G}$ . By the defining Bayes optimality of a strictly proper loss,  $F_k^*(\mathcal{M}')$  equals the Bayes risk  $\mathcal{R}(\mathcal{G}')$  attained by the  $\sigma(X_{\mathcal{M}'})$ -measurable Bayes act. The Bayes risk is non-increasing under refinement of the conditioning  $\sigma$ -field (the tower property: conditioning on a finer field can only reduce the expected divergence), hence  $\mathcal{R}(\mathcal{G}_k) \geq \mathcal{R}(\mathcal{G})$ , which is Eq. (23). For a strictly proper loss, the inequality is strict exactly when the Bayes act associated with the finer field differs from that associated with the coarser field on a positive-probability event.  $\square$

Lemma B.3 explains why a single consensus  $\theta_g$  is corrosive: richer-modality clients drive  $\theta_g$  toward full-modality minimizers, whose risk  $F_k^*(\mathcal{M})$  a modality-limited client cannot attain, so consensus forces the latter to forfeit personalization for an unattainable target.

### B.3. Proof of Proposition 2.1 (Structural Divergence of Local Optima)

We give the argument for squared loss to make the Bayes-optimal step explicit; the conclusion extends to any strictly proper loss by the same conditional-functional argument used in Lemma B.3.

**Assumption B.4** (Bayes-Identifiable Parameterization). For the shared parameterization used in Proposition 2.1, distinct modality-conditioned Bayes-optimal predictors have disjoint sets of population-risk minimizers. In particular, if the Bayes acts associated with  $\mathcal{M}_i$  and  $\mathcal{M}_j$  differ on a set of positive probability, then

$$\arg \min_{\theta \in \Theta} F_i(\theta; \mathcal{M}_i) \cap \arg \min_{\theta \in \Theta} F_j(\theta; \mathcal{M}_j) = \emptyset.$$

**Proposition B.5** (Structural Divergence of Local Optima (Expanded)). *Under Assumption B.4, let  $X_{\mathcal{M}}$  and  $Y$  have joint law  $P$ , and let clients  $i$  and  $j$  share this full-modality law but observe  $\mathcal{M}_i \neq \mathcal{M}_j$ . Under squared loss, define*

$$F(\theta; \mathcal{M}') \triangleq \mathbb{E} \left[ (f_{\theta}(X_{\mathcal{M}'}) - Y)^2 \right],$$

together with

$$\mathcal{G}_i \triangleq \sigma(X_{\mathcal{M}_i}), \quad \mathcal{G}_j \triangleq \sigma(X_{\mathcal{M}_j}).$$

and

$$f_i^* \triangleq \mathbb{E}[Y \mid \mathcal{G}_i], \quad f_j^* \triangleq \mathbb{E}[Y \mid \mathcal{G}_j].$$

If

$$\mathbb{P}(f_i^* \neq f_j^*) > 0,$$

then

$$\arg \min_{\theta \in \Theta} F(\theta; \mathcal{M}_i) \cap \arg \min_{\theta \in \Theta} F(\theta; \mathcal{M}_j) = \emptyset.$$

*Proof.* Under squared loss,  $f_i^*$  and  $f_j^*$  are the Bayes-optimal predictors associated with the two observation  $\sigma$ -fields. If  $\mathcal{M}_i \subsetneq \mathcal{M}_j$ , then the tower property gives

$$\mathbb{E}[f_j^* \mid \mathcal{G}_i] = \mathbb{E}[\mathbb{E}[Y \mid \mathcal{G}_j] \mid \mathcal{G}_i] = \mathbb{E}[Y \mid \mathcal{G}_i] = f_i^*,$$

so  $f_i^*$  is the coarser-information projection of  $f_j^*$ . For non-nested modality sets, the predictor difference is imposed directly by  $\mathbb{P}(f_i^* \neq f_j^*) > 0$ . Assumption B.4 then implies that the corresponding parameter-space minimizer sets are disjoint.  $\square$

Thus, different observation  $\sigma$ -fields can induce different Bayes-optimal predictors and, under Assumption B.4, disjoint parameter-space minimizer sets.

### B.4. Proof of Proposition 2.3 (Incompatible Update Directions)

This subsection provides the proof omitted from the main text, completing the symmetry: three formal claims (Proposition 2.1—2.7), three proofs. Whereas Proposition 2.1 shows the targets diverge, here we show that the Euclidean coupling of Eq. (1) injects update directions that client  $i$  cannot even identify, producing negative transfer.

**Setup.** For the negative log-likelihood loss underlying Eq. (8), define the per-sample score and population gradient by

$$\begin{aligned} s_k(\theta; \mathbf{x}, y) &\triangleq \nabla_{\theta} \log p(y \mid \mathbf{x}_{\mathcal{M}_k}; \theta), \\ g_k(\theta) &\triangleq \nabla_{\theta} F_k(\theta; \mathcal{M}_k) = -\mathbb{E}_{\mathcal{D}_k \mid \mathcal{M}_k} [s_k(\theta; \mathbf{x}, y)]. \end{aligned} \tag{24}$$

and the Fisher and its identifiable subspace

$$\mathbf{F}_k(\theta) \triangleq \mathbb{E}_{\mathcal{D}_k \mid \mathcal{M}_k} [s_k s_k^{\top}], \quad \mathcal{S}_k(\theta) \triangleq \text{range } \mathbf{F}_k(\theta). \tag{25}$$

We use the modality-factorized realization of  $\Theta_k$  adopted in Section 3: the parameter splits as  $\theta = (\theta^{\text{shr}}, \{\theta^{(m)}\}_{m \in \mathcal{M}})$ , where  $\theta^{(m)}$  influences the output only through modality  $m$ 's input branch, and the absent blocks  $\{m \notin \mathcal{M}_k\}$  are inactive for client  $k$ . Let  $\mathcal{A}_k$  index the active coordinates of  $k$  (i.e.,  $\theta^{\text{shr}}$  and  $\{\theta^{(m)} : m \in \mathcal{M}_k\}$ ) and  $E_{\mathcal{A}_k} \triangleq \text{span}\{e_a : a \in \mathcal{A}_k\}$ .

We assume the mild *marginal consistency* of a shared backbone: masking the inputs of  $\mathcal{M}_j \setminus \mathcal{M}_i$  reduces client  $j$ 's model to client  $i$ 's, so

$$s_i(\theta; \mathbf{x}_{\mathcal{M}_i}, y) = \Pi_{\mathcal{A}_i} s_j(\theta; \mathbf{x}_{\mathcal{M}_j}, y) \Big|_{\mathcal{M}_j \setminus \mathcal{M}_i \text{ masked}}, \quad (26)$$

where  $\Pi_{\mathcal{A}_i}$  is the coordinate projector onto  $E_{\mathcal{A}_i}$ . This is exactly the modality-dropout-consistent, shared-encoder parameterization used by the method.

**Assumption B.6** (Nested Block-Fisher Support). Under the modality-factorized parameterization, the local Fisher information is block diagonal, is positive definite on every active shared or modality-specific block, and is zero on inactive modality blocks. Hence,

$$\text{range } \mathbf{F}_k(\theta) = E_{\mathcal{A}_k},$$

i.e., its range equals the active coordinate subspace induced by the client's modality set.

**Lemma B.7** (Score Support and Gradient Containment). *For every  $k$  and  $\theta$ : (a)  $s_k(\theta; \cdot) \in E_{\mathcal{A}_k}$  almost surely, hence  $\mathcal{S}_k(\theta) \subseteq E_{\mathcal{A}_k}$ ; (b)  $\mathbf{g}_k(\theta) \in \mathcal{S}_k(\theta)$ .*

*Proof.* (a) Coordinates outside  $\mathcal{A}_k$  index inactive branches, so  $f_\theta(\mathbf{x}_{\mathcal{M}_k})$  does not depend on them; their entries of  $s_k$  vanish a.s., giving  $s_k \in E_{\mathcal{A}_k}$  and therefore  $\mathbf{F}_k = \mathbb{E}[s_k s_k^\top]$  has range in  $E_{\mathcal{A}_k}$ . (b) Since  $\mathcal{S}_k = \text{range } \mathbf{F}_k = \overline{\text{span}} \text{supp}(s_k)$ , each realization satisfies  $s_k \in \mathcal{S}_k$ ; as  $\mathcal{S}_k$  is a closed subspace,  $\mathbf{g}_k = -\mathbb{E}[s_k] \in \mathcal{S}_k$ .  $\square$

**Proposition B.8** (Incompatible Update Directions (Expanded)). *Under Assumption B.6, let  $\mathcal{M}_i \subsetneq \mathcal{M}_j$ . Then at every  $\theta$ :*

- (i) **Nesting:**  $\mathcal{S}_i(\theta) \subsetneq \mathcal{S}_j(\theta)$  and  $\mathcal{S}_j(\theta) \cap \mathcal{S}_i(\theta)^\perp \neq \{0\}$ .
- (ii) **First-order unidentifiability:** every  $v \in \mathcal{S}_i(\theta)^\perp$  satisfies  $v^\top \mathbf{F}_i(\theta) v = 0$  and  $\langle \mathbf{g}_i(\theta), v \rangle = 0$ ; i.e.,  $v$  is invisible to both the curvature and the first-order objective of client  $i$ .
- (iii) **Conflicted proximal pull:** If  $P_{\mathcal{S}_i(\theta)^\perp}(\theta_g - \theta_i) \neq 0$ , then the proximal force in Eq. (1) contains a locally unidentifiable component that produces no first-order decrease of  $F_i$ .

*Proof.* (i) By Assumption B.6,  $\mathcal{S}_i(\theta) = E_{\mathcal{A}_i}$ ,  $\mathcal{S}_j(\theta) = E_{\mathcal{A}_j}$ . Since  $\mathcal{M}_i \subseteq \mathcal{M}_j$ , the corresponding active coordinate sets satisfy  $\mathcal{A}_i \subseteq \mathcal{A}_j$ , and hence  $\mathcal{S}_i(\theta) = E_{\mathcal{A}_i} \subseteq E_{\mathcal{A}_j} = \mathcal{S}_j(\theta)$ . Moreover, if  $\mathcal{M}_i \subsetneq \mathcal{M}_j$ , then  $\mathcal{A}_i \subsetneq \mathcal{A}_j$ . Therefore, any nonzero  $v \in E_{\mathcal{A}_j \setminus \mathcal{A}_i} \triangleq \text{span}\{e_a : a \in \mathcal{A}_j \setminus \mathcal{A}_i\}$  satisfies  $v \in \mathcal{S}_j(\theta) \cap \mathcal{S}_i(\theta)^\perp$ . Consequently,  $\mathcal{S}_j(\theta) \cap \mathcal{S}_i(\theta)^\perp \neq \{0\}$ , and the inclusion  $\mathcal{S}_i(\theta) \subsetneq \mathcal{S}_j(\theta)$  is strict.

(ii) For  $v \in \mathcal{S}_i^\perp$ ,  $v^\top \mathbf{F}_i v = \mathbb{E}[(s_i^\top v)^2] = 0$  because  $s_i \in \mathcal{S}_i$  and  $s_i^\top v = 0$  a.s.; the same almost-sure orthogonality gives  $\langle \mathbf{g}_i, v \rangle = -\mathbb{E}[s_i^\top v] = 0$ , and indeed  $\mathbf{g}_i \in \mathcal{S}_i$  by Lemma B.7(b).

(iii) Decompose  $\theta_g - \theta_i = P_{\mathcal{S}_i}(\theta_g - \theta_i) + P_{\mathcal{S}_i^\perp}(\theta_g - \theta_i)$ . If the second component is nonzero, then by (ii) it is orthogonal to the local first-order signal and lies in the null Fisher subspace. It therefore produces no first-order decrease of  $F_i$ .  $\square$

**Remark B.9** (Negative Transfer via Conflicted Pull: Variance Inflation Mechanism). Under a fixed-subspace local linearization, consider client  $i$  running stochastic descent on  $F_i(\theta; \mathcal{M}_i) + \frac{\lambda}{2} \|\theta - \theta_g\|^2$ , with  $\theta_g$  fixed during the local recursion. Let  $\{\xi^t\}$  be additional i.i.d. zero-mean perturbations with covariance  $\Sigma$ , not assumed to be supported on  $\mathcal{S}_i$ , and suppose  $0 < \eta\lambda < 2$ . Projecting onto  $\mathcal{S}_i^\perp$  with  $P \triangleq P_{\mathcal{S}_i^\perp}$ , and using Proposition B.8(ii), gives  $P \mathbf{g}_i = 0$ . Hence the unidentifiable-subspace iterate  $u^t \triangleq P(\theta_i^t - \theta_g)$  obeys

$$u^{t+1} = (1 - \eta\lambda)u^t - \eta P \xi^t, \quad \mathbb{E}[\xi^t] = 0, \quad \text{Cov}(\xi^t) = \Sigma. \quad (27)$$

Its stationary second moment is  $\mathbb{E}\|u^\infty\|^2 = \frac{\eta}{2\lambda - \eta\lambda^2} \text{tr}(P\Sigma P)$ , which is positive whenever  $\text{tr}(P\Sigma P) > 0$ . Thus, directions carrying zero local first-order signal can retain nonzero stationary variance under the Euclidean consensus recursion. This provides one possible variance-inflation mechanism for negative transfer and motivates the overlap-restricted coupling  $d_{\mathcal{Z}}^2(\pi_{\mathcal{S}_{ij}}(z_i), \pi_{\mathcal{S}_{ij}}(z_j))$  in Eq. (2), which acts only on shared, mutually identifiable supports.

### B.5. Well-Posedness of the Edge Couplings

When  $\mathcal{M}_i \neq \mathcal{M}_j$ , the two landscapes live on incompatible product spaces; the joint role of  $\mathcal{T}_k$  and  $\pi_S$  is to construct one comparable representation per edge, so that each distance in Eqs. (2)–(3) is well-defined.

**Lemma B.10** (Edge-wise Comparability via Restriction). *Fix  $k \neq j$  with  $S_{kj} \neq \emptyset$ . Under Assumption 2.6(ii), the restricted representations*

$$\pi_{S_{kj}}(\mathcal{T}_k(\theta_k)), \pi_{S_{kj}}(\mathcal{T}_j(\theta_j)) \in \mathcal{Z}_{S_{kj}} \quad (28)$$

*lie in the same metric space  $(\mathcal{Z}_{S_{kj}}, d_{\mathcal{Z}})$  for all  $\theta_k \in \Theta_k, \theta_j \in \Theta_j$ , so the edge distance  $d_{\mathcal{Z}}^2(\pi_{S_{kj}}(z_k), \pi_{S_{kj}}(z_j))$  is well-defined and symmetric in  $(k, j)$ . If  $S_{kj} = \emptyset$  then  $\omega_{kj} = 0$  (Assumption 2.6(iii)) and the edge contributes no term, so no undefined restriction is ever evaluated.*

*Proof.* By Assumption 2.6(ii), the transformations  $\mathcal{T}_k$  and  $\mathcal{T}_j$  map the local parameters into a common ambient space  $\mathcal{Z}$ , and the restriction  $\pi_{S_{kj}}$  projects onto the shared coordinate block  $\mathcal{Z}_{S_{kj}}$ . Consequently,  $\pi_{S_{kj}}(\mathcal{T}_k(\theta_k))$  and  $\pi_{S_{kj}}(\mathcal{T}_j(\theta_j))$  both lie in the same metric space  $(\mathcal{Z}_{S_{kj}}, d_{\mathcal{Z}})$ , rendering the edge distance  $d_{\mathcal{Z}}^2(\cdot, \cdot)$  well-defined and symmetric in  $(k, j)$ . Assumption 2.6(iii) guarantees that whenever  $S_{kj} = \emptyset$ , the corresponding weight  $\omega_{kj} = 0$ , so the edge contributes no term and no undefined restriction is ever evaluated. The symmetry follows because both endpoints are processed by the same operator  $\pi_{S_{kj}}$  on the undirected edge  $\{k, j\}$  and  $d_{\mathcal{Z}}$  is a metric.  $\square$

Lemma B.10 is the structural prerequisite for the potential property below: it guarantees that the term shared between  $J_k$  and  $J_j$  on edge  $\{k, j\}$  is one and the same quantity.

### B.6. Proof of Proposition 2.7 (Exact Potential Game)

**Proposition B.11** (Exact Potential Game (Expanded)). *Consider the payoffs  $J_k$  of Eq. (2) and the candidate potential  $V$  of Eq. (3). Assume (i)  $\omega_{kj} = \omega_{jk}$  for all  $k \neq j$ ; (ii) the same restriction  $\pi_{S_{kj}}$  is used on each undirected edge  $\{k, j\}$ ; (iii)  $\omega_{kj} > 0 \iff S_{kj} \neq \emptyset$ . Then  $V$  is an exact potential: for every  $k$ , every fixed  $W_{-k}$ , and all  $\theta_k, \theta'_k \in \Theta_k$ ,*

$$J_k(\theta'_k; W_{-k}) - J_k(\theta_k; W_{-k}) = V(\theta'_k, W_{-k}) - V(\theta_k, W_{-k}). \quad (29)$$

*Proof.* Define the edge discrepancy, well-defined whenever  $\omega_{kj} > 0$  by Lemma B.10,

$$D_{kj}(\theta_k, \theta_j) \triangleq d_{\mathcal{Z}}^2(\pi_{S_{kj}}(\mathcal{T}_k(\theta_k)), \pi_{S_{kj}}(\mathcal{T}_j(\theta_j))), \quad D_{kj} = D_{jk}. \quad (30)$$

Fix  $k$  and hold  $W_{-k}$  constant. Split the potential into the  $k$ -local term, the other clients' local terms, and the pairwise sum:

$$V(W) = \underbrace{w_k F_k(\theta_k; \mathcal{I}_k)}_{\text{depends on } \theta_k} + \underbrace{\sum_{i \neq k} w_i F_i(\theta_i; \mathcal{I}_i)}_{\text{constant in } \theta_k} + \frac{\lambda}{2} \sum_{1 \leq i < j \leq K} \omega_{ij} D_{ij}(\theta_i, \theta_j). \quad (31)$$

Partition the unordered-pair sum into edges incident to  $k$  and edges avoiding  $k$ :

$$\sum_{1 \leq i < j \leq K} \omega_{ij} D_{ij} = \sum_{j \neq k} \omega_{\min\{k, j\} \max\{k, j\}} D_{kj}(\theta_k, \theta_j) + \sum_{\substack{1 \leq i < j \leq K \\ i \neq k, j \neq k}} \omega_{ij} D_{ij}(\theta_i, \theta_j). \quad (32)$$

Because the edge set is undirected,  $\omega_{kj} = \omega_{jk}$ , and  $\omega_{kj} = 0$  for non-neighbors, the first sum collapses to  $\sum_{j \in \mathcal{N}_k} \omega_{kj} D_{kj}(\theta_k, \theta_j)$ , while the second is constant in  $\theta_k$ . Hence

$$V(W) = \left( w_k F_k(\theta_k; \mathcal{I}_k) + \frac{\lambda}{2} \sum_{j \in \mathcal{N}_k} \omega_{kj} D_{kj}(\theta_k, \theta_j) \right) + \Phi(W_{-k}), \quad (33)$$

where  $\Phi(W_{-k})$  gathers every term independent of  $\theta_k$ . The bracket is exactly  $J_k(\theta_k; W_{-k})$  of Eq. (2). Subtracting the values at  $\theta'_k$  and  $\theta_k$  cancels  $\Phi(W_{-k})$  and yields Eq. (29).  $\square$

The factor  $\frac{1}{2}$  in  $V$  and the once-per-unordered-pair summation are exactly what convert the per-client neighbor sums of  $\{J_k\}$  into a single shared function; symmetry  $\omega_{kj} = \omega_{jk}$  together with the common  $\pi_{S_{kj}}$  (Lemma B.10) is what makes the cancellation exact rather than approximate. Consequently, every local minimizer of  $V$  admits no profitable unilateral deviation and is a local Nash equilibrium, as stated below Eq. (4).

### B.7. Gradient-Flow Stability: Lyapunov Descent and LaSalle Convergence

We discharge the stability claims attached to the Riemannian gradient flow Eq. (5), namely the dissipation identity Eq. (6) and convergence to first-order Nash equilibria.

**Definition B.12** (First-Order Nash Equilibrium). Assume each  $\Theta_k$  is a smooth manifold (or a retraction-compatible constraint set) and each  $J_k(\cdot; W_{-k})$  is differentiable. A profile  $W^* \in \mathcal{W}$  is a *first-order Nash equilibrium* (FONE) if  $\text{grad}_{g_k, \theta_k} J_k(\theta_k^*; W_{-k}^*) = 0$  for every  $k$ .

**Proposition B.13** (Lyapunov Property of  $V$  and LaSalle Convergence). *Let  $V$  be the continuously differentiable exact potential of Proposition B.11 on the product manifold  $(\mathcal{W}, g)$  with  $g \triangleq \oplus_k g_k$ , and let  $W(\cdot)$  solve the flow Eq. (5),  $\dot{\theta}_k = -\text{grad}_{g_k, \theta_k} V(W)$ . Then along any solution,*

$$\frac{d}{dt} V(W(t)) = - \sum_{k=1}^K \|\text{grad}_{g_k, \theta_k} V(W(t))\|_{g_k}^2 \leq 0, \quad (34)$$

with equality iff  $\text{grad} V(W(t)) = 0$ . By Lemma B.14, this is equivalent to  $W(t)$  being a FONE. If moreover the sublevel set  $\{W : V(W) \leq V(W(0))\}$  is compact, then every solution is bounded and its  $\omega$ -limit set is a nonempty, connected, invariant subset of the FONE set; hence  $\text{grad} V(W(t)) \rightarrow 0$  and  $W(t)$  converges to the set of first-order Nash equilibria, irrespective of the clients' geometric heterogeneity.

*Proof.* On the product manifold the gradient decomposes blockwise,  $\text{grad} V = (\text{grad}_{g_k, \theta_k} V)_k$ , and  $\langle \cdot, \cdot \rangle_g = \sum_k \langle \cdot, \cdot \rangle_{g_k}$ . By the chain rule along the flow,

$$\frac{d}{dt} V(W(t)) = \sum_k \langle \text{grad}_{g_k, \theta_k} V, \dot{\theta}_k \rangle_{g_k} = - \sum_k \langle \text{grad}_{g_k, \theta_k} V, \text{grad}_{g_k, \theta_k} V \rangle_{g_k} = - \sum_k \|\text{grad}_{g_k, \theta_k} V\|_{g_k}^2, \quad (35)$$

which is Eq. (34); the sum vanishes iff every block gradient vanishes, *i.e.*  $W(t)$  is a FONE. Under compactness of the sublevel set, monotonicity of  $V$  along the flow keeps  $W(t)$  in that compact set, so the solution is bounded and  $V(W(t))$  decreases to a finite limit. LaSalle's invariance principle then forces the  $\omega$ -limit set into the largest invariant set contained in  $\{W : \frac{d}{dt} V = 0\} = \{W : \text{grad} V(W) = 0\}$ , which is exactly the FONE set, giving  $\text{grad} V(W(t)) \rightarrow 0$  and convergence to that set.  $\square$

### B.8. Gradient Alignment and the Decentralized Realization

Finally, we justify the ‘‘Decentralized Realization’’ claim that each client may descend its *own* payoff  $J_k$  while collectively minimizing  $V$ .

**Lemma B.14** (Blockwise Gradient Alignment). *If the game is an exact potential game with a differentiable potential  $V$  (Proposition B.11), then for every  $k$  and every  $W \in \mathcal{W}$ ,*

$$\text{grad}_{g_k, \theta_k} J_k(\theta_k; W_{-k}) = \text{grad}_{g_k, \theta_k} V(W). \quad (36)$$

Consequently,  $W^*$  is a first-order stationary point of  $V$  ( $\text{grad}_{g_k, \theta_k} V(W^*) = 0$  for all  $k$ ) if and only if  $W^*$  is a FONE.

*Proof.* By the exact-potential identity Eq. (29),  $J_k(\cdot; W_{-k})$  and  $V(\cdot, W_{-k})$  differ by the additive constant  $\Phi(W_{-k})$  that is independent of  $\theta_k$ . Taking the directional derivative along any tangent direction  $\xi_k \in T_{\theta_k} \Theta_k$  therefore gives identical directional derivatives in the  $k$ -th block; by Riesz representation under the metric  $g_k$  the Riemannian gradients coincide. The stated equivalence is then immediate from Definition B.12.  $\square$

Lemma B.14 is the formal content of the main text's decentralized dynamics: replacing  $\text{grad}_{g_k, \theta_k} V$  by  $\text{grad}_{g_k, \theta_k} J_k$  in Eq. (5) leaves the flow—and hence the Lyapunov descent of Proposition B.13—unchanged, so each client descending its local payoff monotonically decreases the global potential  $V$  without explicit coordination on  $V$ .

## C. Appendix for the Methodology

This appendix discharges the mechanical claims of Section 3 for both solvers: the idealized *reference solver* (Part C) and its communication-feasible realization *GeoEvo* (Part C.3). We establish per-round, per-client, deterministic structural properties: the identity-aware geometry of the descent, the well-posedness of every operator, and a proven majorization that bridges the global potential  $V$  and the local prototype surrogate.

**Notation.**  $z_k = \mathcal{T}_k(\theta_k) \in \mathcal{Z}$ ;  $\pi_S : \mathcal{Z} \rightarrow \mathcal{Z}_S$  restricts to support  $S \subseteq \mathcal{M}$ ;  $S_{kj} = \mathcal{M}_k \cap \mathcal{M}_j$ ;  $\mathcal{N}_k^t = \mathcal{N}_k \cap \mathcal{S}^t$  are the participating neighbors at round  $t$ ;  $S_k^t = \bigcup_{j \in \mathcal{N}_k^t} S_{kj} \subseteq \mathcal{M}_k$  and  $\tilde{\omega}_k^t = \sum_{j \in \mathcal{N}_k^t} \omega_{kj}$ ;  $\tilde{\mathbf{F}}_k(\theta_k) = \mathbf{F}_k(\theta_k) + \epsilon \mathbf{I}$  is the damped Fisher metric (Remark 3.1);  $\text{grad}_{g_k, \theta_k}$  is the Riemannian gradient under  $g_k$ .

## Part I. Reference Solver: Exact Potential Minimization

### C.1. Identity-Consistent Fisher Metric and Its Block-Sparsity

The main text equips each  $\Theta_k$  with the damped empirical-Fisher metric of Eq. (8) and asserts that it “inherits the block-sparse structure of  $\mathcal{M}_k$  without explicit masking.” We make this precise. Adopt the modality-factorized realization of Section 3:  $\theta_k = (\theta_k^{\text{shr}}, \{\theta_k^{(m)}\}_{m \in \mathcal{M}})$ , where  $\theta_k^{(m)}$  acts on the output only through modality  $m$ ’s input branch, and for client  $k$  the branches  $m \notin \mathcal{M}_k$  are inactive.

**Lemma C.1** (Score Sparsity  $\Rightarrow$  Fisher Block-Sparsity). *Let  $s_k(\theta_k; \mathbf{x}, y) \triangleq \nabla_{\theta_k} \log p(y \mid \mathbf{x}_{\mathcal{M}_k}; \theta_k)$  be the score. For every coordinate block  $m \notin \mathcal{M}_k$ , the partial score  $\nabla_{\theta_k^{(m)}} \log p(y \mid \mathbf{x}_{\mathcal{M}_k}; \theta_k) \equiv 0$ . Consequently the empirical Fisher  $\mathbf{F}_k(\theta_k) = \frac{1}{|\mathcal{D}_k|} \sum s_k s_k^\top$  has zero rows and columns on all blocks  $m \notin \mathcal{M}_k$ ; i.e.,  $\mathbf{F}_k$  is supported on the active index set  $\mathcal{A}_k = \{\theta_k^{\text{shr}}\} \cup \{\theta_k^{(m)} : m \in \mathcal{M}_k\}$ . The damped metric  $\tilde{\mathbf{F}}_k = \mathbf{F}_k + \epsilon \mathbf{I}$  is therefore block-diagonal across the active/inactive split, so its inverse  $\tilde{\mathbf{F}}_k^{-1}$  preserves this split (acting as  $\frac{1}{\epsilon} \mathbf{I}$  on the inactive block); since the descent objective ( $F_k$  and  $\mathcal{T}_k$ ) does not depend on the inactive coordinates,  $\nabla_{\theta_k} V$  vanishes there and the preconditioned direction  $\tilde{\mathbf{F}}_k^{-1} \nabla_{\theta_k} V$  stays supported on  $\mathcal{A}_k$ .*

*Proof.* Since branch  $m \notin \mathcal{M}_k$  is inactive,  $p(y \mid \mathbf{x}_{\mathcal{M}_k}; \theta_k)$  does not depend on  $\theta_k^{(m)}$ , so  $\partial_{\theta_k^{(m)}} \log p \equiv 0$  pointwise; hence those entries of  $s_k$  vanish for every sample. An outer product  $s_k s_k^\top$  with zero entries on block  $m$  has zero rows or columns there, and the property is preserved under averaging; adding  $\epsilon \mathbf{I}$  keeps  $\tilde{\mathbf{F}}_k$  block-diagonal across the active/inactive split, so  $\tilde{\mathbf{F}}_k^{-1}$  does not mix the two blocks. Restricting the positive-definite damped matrix  $\tilde{\mathbf{F}}_k$  to  $T_{\theta_k} \Theta_k$  yields the metric  $g_k$ .  $\square$

Lemma C.1 is what makes the metric *identity-consistent*: the geometry never spends curvature on modalities that the client cannot observe, so the natural-gradient preconditioner  $\tilde{\mathbf{F}}_k^{-1}$  acts only on  $\mathcal{M}_k$ -relevant coordinates, matching the structural identity of Definition 2.4.

### C.2. Block Descent as Steepest Descent under the Fisher Metric

We justify the preconditioned step shared by Eq. (9) (reference solver) and Eq. (13) (GeoEvo). All matrix expressions below are written in local coordinates on  $T_{\theta_k} \Theta_k$ ; equivalently, an embedded realization uses the corresponding tangent-space natural gradient.

**Lemma C.2** (Metric Steepest Descent). *Let  $h : \Theta_k \rightarrow \mathbb{R}$  be differentiable,  $g_k(\theta_k) \succ 0$  a metric, and  $\eta > 0$ . The trust-region model*

$$\Delta^* = \arg \min_{\Delta \in T_{\theta_k} \Theta_k} \langle \nabla_{\theta_k} h(\theta_k), \Delta \rangle + \frac{1}{2\eta} \|\Delta\|_{g_k(\theta_k)}^2 \quad (37)$$

*has the closed-form minimizer  $\Delta^* = -\eta g_k(\theta_k)^{-1} \nabla_{\theta_k} h(\theta_k) = -\eta \text{grad}_{g_k, \theta_k} h(\theta_k)$ , and the retracted update  $\theta_k^+ = \text{Retr}_{\theta_k}(\Delta^*)$  is a first-order approximation of the Riemannian gradient flow  $\dot{\theta}_k = -\text{grad}_{g_k, \theta_k} h$ .*

*Proof.* Eq. (37) is a strictly convex quadratic in  $\Delta$  (as  $g_k \succ 0$ ); its stationarity condition  $\nabla h + \frac{1}{\eta} g_k \Delta = 0$  gives  $\Delta^* = -\eta g_k^{-1} \nabla h$ . By definition of the Riemannian gradient,  $g_k^{-1} \nabla h = \text{grad}_{g_k, \theta_k} h$ . A retraction satisfies  $\text{Retr}_{\theta_k}(\tau \xi) = \theta_k + \tau \xi + \mathcal{O}(\tau^2)$ , so  $\theta_k^+ = \theta_k - \eta \text{grad}_{g_k, \theta_k} h + \mathcal{O}(\eta^2)$ , i.e. an explicit-Euler step of the gradient flow.  $\square$

Instantiating  $g_k = \tilde{\mathbf{F}}_k$  in Lemma C.2 yields the natural-gradient direction  $-\eta \tilde{\mathbf{F}}_k^{-1} \nabla_{\theta_k} h$ , which is steepest descent measured in the model’s KL geometry. With  $h = V$  this is exactly Eq. (9); with  $h = \tilde{J}_k^t$  (Part C.3) it is Eq. (13). By Lemma C.1, the direction is automatically supported on  $\mathcal{A}_k$ , so no explicit masking is required.

### C.3. Per-Step Sufficient Decrease

We prove Lemma 3.2 in the *simultaneous* form used throughout the paper: at each iteration *all* blocks move along their own Riemannian gradient,  $W^{t+1} = (\text{Retr}_{\theta_k^t}(-\eta_t \text{grad}_{g_k, \theta_k} V(W^t)))_{k=1}^K$ . This matches Eq. (9) and avoids any single-block serial convention.

**Lemma C.3** (Per-step Sufficient Decrease (Simultaneous Blocks)). *Let  $V$  be  $L$ -smooth w.r.t. the retraction on the product manifold  $(\mathcal{W}, g)$ ,  $g = \oplus_k g_k$  (Assumption 4.2). For the simultaneous update with  $0 < \eta_t \leq 1/L$ ,*

$$V(W^{t+1}) \leq V(W^t) - \frac{\eta_t}{2} \|\text{grad } V(W^t)\|_g^2 = V(W^t) - \frac{\eta_t}{2} \sum_{k=1}^K \|\text{grad}_{g_k, \theta_k} V(W^t)\|_{g_k}^2. \quad (38)$$

*Proof.* The simultaneous step is the single product-manifold tangent vector  $\xi = -\eta_t \text{grad } V(W^t) \in T_{W^t} \mathcal{W}$ , with  $W^{t+1} = \text{Retr}_{W^t}(\xi)$ . The product structure gives the orthogonal block decomposition  $\|\xi\|_g^2 = \sum_k \|\xi_k\|_{g_k}^2$  and  $\langle \text{grad } V, \xi \rangle_g = \sum_k \langle \text{grad}_{g_k, \theta_k} V, \xi_k \rangle_{g_k}$ . Applying the  $L$ -smooth descent inequality of Assumption 4.2,

$$V(W^{t+1}) \leq V(W^t) + \langle \text{grad } V(W^t), \xi \rangle_g + \frac{L}{2} \|\xi\|_g^2 = V(W^t) - \left( \eta_t - \frac{L\eta_t^2}{2} \right) \|\text{grad } V(W^t)\|_g^2. \quad (39)$$

For  $\eta_t \leq 1/L$  we have  $\eta_t - \frac{L\eta_t^2}{2} \geq \frac{\eta_t}{2}$ , giving Eq. (38); the second equality is the block decomposition of the product norm.  $\square$

**Corollary C.4** (Telescoped Stationarity of the Reference Solver). *Summing Eq. (38) and using  $V \geq V_{\inf}$  (Assumption 4.3(i)),*

$$\sum_{t=0}^{\infty} \frac{\eta_t}{2} \|\text{grad } V(W^t)\|_g^2 \leq V(W^0) - V_{\inf} < \infty. \quad (40)$$

Hence if  $\inf_t \eta_t > 0$  then  $\|\text{grad } V(W^t)\|_g \rightarrow 0$ , so  $\|\text{grad}_{g_k, \theta_k} V(W^t)\|_{g_k} \rightarrow 0$  for every  $k$ .

## Part II. GeoEvo: Federated Geometric Evolution

### C.4. Masked Structure Space and a Monotonicity Lemma

We instantiate  $\mathcal{Z} = \bigoplus_{m \in \mathcal{M}} \mathcal{Z}^{(m)}$ , where  $\mathcal{Z}^{(m)} = \mathbb{R}^{q_m}$ , let  $\pi_S$  retain the blocks indexed by  $S$ , and define  $d_{\mathcal{Z}}$  coordinatewise:

$$d_{\mathcal{Z}}^2(\pi_S(z), \pi_S(z')) = \sum_{m \in S} \|z^{(m)} - z'^{(m)}\|^2, \quad \pi_S \circ \pi_{S'} = \pi_S \quad (S \subseteq S'). \quad (41)$$

By the convention  $\omega_{kj} > 0 \iff S_{kj} \neq \emptyset$ , no empty-overlap distance is evaluated. When  $\mathcal{N}_k^t = \emptyset$ , we set  $S_k^t = \emptyset$ ,  $\bar{\omega}_k^t = 0$ , and  $Z_k^t = z_k^t$ ; the surrogate coupling then vanishes. The nontrivial prototype analysis below therefore considers  $\mathcal{N}_k^t \neq \emptyset$ .

**Lemma C.5** (Masked-Distance Monotonicity). *For  $S \subseteq S' \subseteq \mathcal{M}$  and any  $z, z' \in \mathcal{Z}$ ,  $d_{\mathcal{Z}}^2(\pi_S(z), \pi_S(z')) \leq d_{\mathcal{Z}}^2(\pi_{S'}(z), \pi_{S'}(z'))$ , with gap  $\sum_{m \in S' \setminus S} \|z^{(m)} - z'^{(m)}\|^2 \geq 0$ .*

*Proof.* Immediate from Eq. (41): enlarging the index set only adds non-negative squared terms.  $\square$

Lemma C.5 formalizes the intuition that evaluating a coupling on the *union* support  $S_k^t$  (rather than on a smaller per-pair overlap  $S_{kj} \subseteq S_k^t$ ) can only *increase* the measured discrepancy—this is the geometric seed of the majorization used in §C.6.

### C.5. Server Prototype: Existence and Coordinatewise Form

The server prototype solves the Fréchet problem Eq. (11):  $Z_k^t \in \arg \min_{z \in \mathcal{Z}} \sum_{j \in \mathcal{N}_k^t} \omega_{kj} d_{\mathcal{Z}}^2(\pi_{S_{kj}}(z), \pi_{S_{kj}}(z_j^t))$ . For each  $m \in S_k^t$  write the neighbors that possess  $m$  and their aggregate weight as

$$\mathcal{N}_k^t(m) \triangleq \{j \in \mathcal{N}_k^t : m \in S_{kj}\}, \quad \Omega_m^t \triangleq \sum_{j \in \mathcal{N}_k^t(m)} \omega_{kj} > 0. \quad (42)$$

**Lemma C.6** (Existence and Coordinatewise Weighted Mean). *Under Eq. (41), the Fréchet objective separates across coordinates, the minimizer exists and, on each  $m \in S_k^t$ , is the weighted mean of the participating neighbors,*

$$Z_k^{t,(m)} = \frac{1}{\Omega_m^t} \sum_{j \in \mathcal{N}_k^t(m)} \omega_{kj} z_j^{t,(m)}, \quad m \in S_k^t. \quad (43)$$

*Proof.* Interchanging the finite sums, the objective equals  $\sum_{m \in S_k^t} \sum_{j \in \mathcal{N}_k^t(m)} \omega_{kj} \|z^{(m)} - z_j^{t,(m)}\|^2$ , a sum of independent strictly convex coordinate problems. Each is a weighted least-squares problem minimized at the weighted mean Eq. (43). To make the vector-valued PSO update unambiguous, we use the canonical completion  $Z_k^{t,(m)} = z_k^{t,(m)}$ ,  $m \notin S_k^t$ . Thus, coordinates outside the union support exert no PSO force on client  $k$ .  $\square$

In particular,  $Z_k^t$  minimizes the exact per-pair coupling, recovering the variational optimality stated as Proposition 4.6; the present lemma supplies the constructive coordinatewise form used below.

**Remark C.7** (Per-Round Communication Complexity). Let  $q_m \triangleq \dim \mathcal{Z}^{(m)}$ ,  $q(S) \triangleq \sum_{m \in S} q_m$ ,  $|z| \triangleq q(\mathcal{M})$ . At each round, every participating client  $k \in \mathcal{S}^t$  receives one triple  $(Z_k^t, S_k^t, \bar{\omega}_k^t)$ , consisting of one structural vector of dimension  $q(S_k^t)$ , one support mask, and one scalar weight. Each participating client uploads only  $z_k^{t+1} = \mathcal{T}_k(\theta_k)$ . Hence GeoEvo uses  $\mathcal{O}(|\mathcal{S}^t|) = \mathcal{O}(K)$  messages and  $\mathcal{O}(\sum_{k \in \mathcal{S}^t} q(S_k^t) + |\mathcal{S}^t||z|) = \mathcal{O}(|\mathcal{S}^t||z|)$  transmitted scalars per round, up to lower-order mask metadata, independently of the neighborhood sizes  $\{|\mathcal{N}_k^t|\}$ . By contrast, the reference solver of Part C exchanges a representation along every active edge, requiring  $\sum_k |\mathcal{N}_k^t| = \mathcal{O}(K^2)$  messages and  $\mathcal{O}(K^2|z|)$  transmitted scalars in the dense-coupling regime.

## C.6. Prototype Compression as a Proven Geometric Majorization

This subsection establishes why optimizing the single-prototype surrogate serves as a principled proxy for the true many-neighbor coupling. We prove a majorization inequality that decomposes the approximation gap into two distinct non-negative sources: neighbor dispersion and weight over-counting. Consequently, the deviation between the “per-pair mask vs. union mask” is fully accounted for.

Define the *exact per-pair coupling* (the  $\theta_k$ -dependent block of  $V$  with neighbors frozen at their round- $t$  uploads; matching  $\Phi_k^t$  of Sec. 4) and the *prototype surrogate coupling* (matching  $\tilde{\Phi}_k^t$ , the coupling part of Eq. (12)):

$$\Phi_k^t(z) \triangleq \frac{\lambda}{2} \sum_{j \in \mathcal{N}_k^t} \omega_{kj} d_{\mathcal{Z}}^2(\pi_{S_{kj}}(z), \pi_{S_{kj}}(z_j^t)), \quad (44)$$

$$\tilde{\Phi}_k^t(z) \triangleq \frac{\lambda}{2} \bar{\omega}_k^t d_{\mathcal{Z}}^2(\pi_{S_k^t}(z), \pi_{S_k^t}(Z_k^t)). \quad (45)$$

**Lemma C.8** (Bias–Variance (Parallel-Axis) Decomposition of the Exact Coupling). *With  $Z_k^t$  from Eq. (43),*

$$\Phi_k^t(z) = \underbrace{\frac{\lambda}{2} \sum_{m \in S_k^t} \Omega_m^t \|z^{(m)} - Z_k^{t,(m)}\|^2}_{z\text{-dependent (bias)}} + \underbrace{\frac{\lambda}{2} \sum_{m \in S_k^t} \nu_m^t}_{z\text{-independent (dispersion)} =: C_k^t}, \quad \nu_m^t \triangleq \sum_{j \in \mathcal{N}_k^t(m)} \omega_{kj} \|z_j^{t,(m)} - Z_k^{t,(m)}\|^2. \quad (46)$$

*Proof.* Coordinatewise separation as in Lemma C.6, then the weighted parallel-axis identity  $\sum_j \omega_j \|x - a_j\|^2 = \Omega \|x - \bar{a}\|^2 + \sum_j \omega_j \|a_j - \bar{a}\|^2$  with  $\bar{a} = \Omega^{-1} \sum_j \omega_j a_j$ , applied per coordinate with  $x = z^{(m)}$ ,  $a_j = z_j^{t,(m)}$ ,  $\Omega = \Omega_m^t$ ,  $\bar{a} = Z_k^{t,(m)}$ .  $\square$

**Theorem C.9** (Prototype Majorization). *For every  $z \in \mathcal{Z}$  (hence, in particular, for  $z = \mathcal{T}_k(\theta_k) \in \text{Im}(\mathcal{T}_k)$ ),*

$$\Phi_k^t(z) \leq \tilde{\Phi}_k^t(z) + C_k^t, \quad C_k^t = \frac{\lambda}{2} \sum_{m \in S_k^t} \nu_m^t \geq 0, \quad (47)$$

where  $C_k^t$  is independent of  $z$ . Thus, the union-mask prototype coupling  $\tilde{\Phi}_k^t$  majorizes the exact per-pair coupling  $\Phi_k^t$  up to the neighbor-dispersion constant  $C_k^t$ . The bound is tight for all  $z \in \mathcal{Z}$  if and only if  $\Omega_m^t = \bar{\omega}_k^t$  for every  $m \in S_k^t$ . Moreover,  $\tilde{\Phi}_k^t(z) = \Phi_k^t(z)$  for all  $z \in \mathcal{Z}$  if and only if, additionally,  $C_k^t = 0$ , equivalently, all positive-weight neighbors share the full union support and agree on every shared coordinate.

*Proof.* By Lemma C.8,

$$\Phi_k^t(z) - C_k^t = \frac{\lambda}{2} \sum_{m \in S_k^t} \Omega_m^t \|z^{(m)} - Z_k^{t,(m)}\|^2.$$

Since

$$\Omega_m^t = \sum_{j: m \in S_{kj}} \omega_{kj} \leq \sum_{j \in \mathcal{N}_k^t} \omega_{kj} = \bar{\omega}_k^t,$$

we obtain

$$\begin{aligned} & \tilde{\Phi}_k^t(z) + C_k^t - \Phi_k^t(z) \\ &= \frac{\lambda}{2} \sum_{m \in S_k^t} (\bar{\omega}_k^t - \Omega_m^t) \|z^{(m)} - Z_k^{t,(m)}\|^2 \geq 0, \end{aligned}$$

which proves Eq. (47). This difference vanishes for every  $z$  if and only if  $\Omega_m^t = \bar{\omega}_k^t$  for all  $m \in S_k^t$ . Finally, the unshifted equality  $\tilde{\Phi}_k^t \equiv \Phi_k^t$  additionally requires  $C_k^t = 0$ , yielding the stated characterization.  $\square$

Theorem C.9 establishes the precise relationship between the exact per-pair coupling and its single-prototype surrogate. The gap  $\tilde{\Phi}_k^t - (\Phi_k^t - C_k^t) = \frac{\lambda}{2} \sum_m (\bar{\omega}_k^t - \Omega_m^t) \|z^{(m)} - Z_k^{t,(m)}\|^2$  decomposes into (i) a *weight-over-counting* part  $\bar{\omega}_k^t - \Omega_m^t \geq 0$  (the union mask applies the aggregate weight to coordinates owned by only some neighbors) and (ii) the *dispersion* constant  $C_k^t$  (neighbor disagreement on shared coordinates). Both are non-negative, so  $\tilde{\Phi}_k^t$  is a conservative (majorizing) compression—never an under-estimate of the true coupling beyond the additive constant.

**Proposition C.10** (Surrogate Descent Controls the Exact Per-Pair Objective). *Let  $g_k^t(\theta) \triangleq w_k F_k(\theta; \mathcal{I}_k) + \Phi_k^t(\mathcal{T}_k(\theta))$  be the exact per-pair objective and  $\tilde{J}_k^t(\theta) = w_k F_k(\theta; \mathcal{I}_k) + \tilde{\Phi}_k^t(\mathcal{T}_k(\theta))$  the prototype surrogate (Eq. (12)). If an update satisfies  $\tilde{J}_k^t(\theta_k^+) \leq \tilde{J}_k^t(\theta_k^{\text{cur}})$ , then*

$$g_k^t(\theta_k^+) \leq g_k^t(\theta_k^{\text{cur}}) + \rho_k^t, \quad \rho_k^t \triangleq \frac{\lambda}{2} \sum_{m \in S_k^t} (\bar{\omega}_k^t - \Omega_m^t) \|z_k^{\text{cur},(m)} - Z_k^{t,(m)}\|^2 \geq 0, \quad (48)$$

where  $z_k^{\text{cur}} = \mathcal{T}_k(\theta_k^{\text{cur}})$ . If overlaps are homogeneous ( $S_{kj} = S_k^t$  for all  $j \in \mathcal{N}_k^t$ , hence  $\Omega_m^t = \bar{\omega}_k^t$ ), then  $\rho_k^t = 0$  and surrogate descent is exact descent of  $g_k^t$ .

*Proof.* Using majorization at  $\theta_k^+$ , then surrogate descent, then the decomposition Eq. (46) at  $\theta_k^{\text{cur}}$ :

$$\begin{aligned} g_k^t(\theta_k^+) &\leq w_k F_k(\theta_k^+) + \tilde{\Phi}_k^t(\mathcal{T}_k(\theta_k^+)) + C_k^t = \tilde{J}_k^t(\theta_k^+) + C_k^t \leq \tilde{J}_k^t(\theta_k^{\text{cur}}) + C_k^t \\ &= w_k F_k(\theta_k^{\text{cur}}) + \tilde{\Phi}_k^t(z_k^{\text{cur}}) + C_k^t = g_k^t(\theta_k^{\text{cur}}) + [\tilde{\Phi}_k^t(z_k^{\text{cur}}) + C_k^t - \Phi_k^t(z_k^{\text{cur}})] = g_k^t(\theta_k^{\text{cur}}) + \rho_k^t, \end{aligned}$$

where the bracket equals  $\rho_k^t$  by Lemma C.8 and Theorem C.9.  $\square$

**Remark C.11** (Closure of the  $V$ -to-Surrogate Gap). Proposition C.10 is a genuine majorize–minimize statement: it does not claim surrogate descent equals descent of  $g_k^t$  in general, but bounds the slippage by an explicit residual  $\rho_k^t \geq 0$  that vanishes under homogeneous overlaps. Since  $g_k^t$  is exactly the  $\theta_k$ -block of  $V$  with neighbors frozen at  $\{z_j^t\}$ , this provides the per-round, per-client link to the global potential.

### C.7. Structure-Matching Lift: Well-Posedness and Robustness

The lift Eq. (14) is  $\text{Lift}_k(z; \theta_k) \in \arg \min_{\theta \in \Theta_k} d_{\mathcal{Z}}^2(\mathcal{T}_k(\theta), z)$ .

**Lemma C.12** (Existence of the Lift). *For any target  $z \in \mathcal{Z}$ , suppose that  $\mathcal{T}_k$  is continuous and either  $\Theta_k$  is compact or the function  $h_z(\theta) \triangleq d_{\mathcal{Z}}^2(\mathcal{T}_k(\theta), z)$  has compact sublevel sets on  $\Theta_k$ . Then  $\arg \min_{\theta \in \Theta_k} h_z(\theta) \neq \emptyset$ .*

*Proof.* The function  $h_z$  is continuous because  $\mathcal{T}_k$  and the metric are continuous. If  $\Theta_k$  is compact, the extreme-value theorem gives a minimizer. Otherwise, let  $\{\theta_n\}$  be a minimizing sequence. Since  $h_z \geq 0$ , its tail lies in some compact sublevel set of  $h_z$  and therefore has a convergent subsequence. Continuity of  $h_z$  implies that the limit attains  $\inf_{\theta \in \Theta_k} h_z(\theta)$ .  $\square$

*Remark C.13* (Exact-Lift Existence and Approximate Lifts). The compact-sublevel-set condition is sufficient but not automatic. When  $\mathcal{T}_k$  has bounded range and  $\Theta_k$  is noncompact, the lift objective may fail to be inf-compact, and its infimum need not be attained. Compactness of  $\Theta_k$  is an additional sufficient condition and is not part of Definition 2.4.

Crucially, GeoEvo uses an approximate lift obtained through a few warm-started retraction steps. By Proposition C.14, the monotone selection rule preserves surrogate descent for any candidate returned by this local solver, so the subsequent stationarity guarantee remains valid. Existence of an exact arg min is therefore a property of the idealized lift operator, not a prerequisite for the practical GeoEvo update.

**Proposition C.14** (Approximate Lift Preserves Monotone Selection). *Let  $\theta_k^{\text{NES}}, \theta_k^{\text{PSO}}$  be produced by any lift, e.g. a few warm-started retracted steps from  $\theta_k$ . If the client selects  $\theta_k^+$  by Eq. (17), then  $\tilde{J}_k^t(\theta_k^+) \leq \tilde{J}_k^t(\theta_k^{\text{cur}})$ .*

*Proof.* Selection evaluates the *exact* surrogate  $\tilde{J}_k^t$  on the already-lifted candidates and includes  $\theta_k^{\text{cur}}$  in the pool; the arg min over a set containing  $\theta_k^{\text{cur}}$  cannot exceed  $\tilde{J}_k^t(\theta_k^{\text{cur}})$ , regardless of how the candidates were generated. Inexactness of the lift only affects candidate *quality*, never the monotonicity guarantee.  $\square$

### C.8. NES in Structure Space: Unbiased Smoothed Gradient

With  $\phi_k(z) \triangleq \tilde{J}_k^t(\text{Lift}_k(z; \theta_k))$  and the Gaussian-smoothed objective  $\phi_{k,\sigma_z}(z) \triangleq \mathbb{E}_{\varepsilon \sim \mathcal{N}(0, \mathbf{I})}[\phi_k(z + \sigma_z \varepsilon)]$ , the estimator in Eq. (15) is  $\hat{\nabla}_z \phi_{k,\sigma_z}(z) = \frac{1}{m\sigma_z} \sum_{i=1}^m \phi_k(z + \sigma_z \varepsilon_i) \varepsilon_i$ .

**Lemma C.15** (Unbiasedness of the NES Estimator). *If  $\phi_k$  is measurable with  $\mathbb{E}[|\phi_k(z + \sigma_z \varepsilon)| \|\varepsilon\|] < \infty$ , then  $\mathbb{E}[\hat{\nabla}_z \phi_{k,\sigma_z}(z)] = \nabla_z \phi_{k,\sigma_z}(z)$ .*

*Proof.* The smoothed objective is the Gaussian convolution  $\phi_{k,\sigma}(z) = \int \phi_k(z + \sigma \epsilon) p(\epsilon) d\epsilon$ . Differentiating the Gaussian density under the integral sign and applying integration by parts gives  $\nabla_z \phi_{k,\sigma}(z) = \frac{1}{\sigma} \mathbb{E}_\epsilon[\phi_k(z + \sigma \epsilon) \epsilon]$ . The stated integrability condition justifies differentiation under the integral. Averaging  $m$  independent samples therefore yields the unbiased estimator in Eq. (15).  $\square$

*Remark C.16.* The smoothing radius  $\sigma_z$  trades estimator variance against bias relative to  $\nabla_z \phi_k$  ( $\phi_{k,\sigma_z} \rightarrow \phi_k$  as  $\sigma_z \downarrow 0$ ). NES proposes candidates in the lower-dimensional, identity-aligned space  $\mathcal{Z}$ , enabling moves outside the basin of the exploitation step; monotonicity is never at risk because such proposals are only *accepted* through Eq. (17).

### C.9. PSO in Structure Space: A Proposal Generator

The PSO update Eq. (16) sets the broadcast prototype  $Z_k^t$  as the global attractor, embedding cross-client structural coordination into the swarm.

*Remark C.17* (Role and Safety). We do not require PSO to be a convergent or contractive dynamical system. In GeoEvo, it is purely a *proposal mechanism* in  $\mathcal{Z}$ : candidates are lifted to  $\Theta_k$  and then filtered by the surrogate. By Proposition C.14, any PSO proposal leaves the monotone selection guarantee intact, so global coordination toward  $Z_k^t$  is obtained without imposing stability conditions on the swarm parameters ( $c_0, c_1, c_2$ ).

### C.10. Monotone Selection and Round-wise Surrogate Dissipation

**Proposition C.18** (Surrogate Monotonicity under Selection). *Fix round  $t$  and treat  $(Z_k^t, S_k^t, \bar{\omega}_k^t)$  as constants, so  $\tilde{J}_k^t$  is a fixed function. With  $\theta_k^+$  chosen by Eq. (17) from a pool containing  $\theta_k^{\text{cur}}$ ,  $\tilde{J}_k^t(\theta_k^+) \leq \tilde{J}_k^t(\theta_k^{\text{cur}})$ .*

*Proof.*  $\theta_k^+$  minimizes  $\tilde{J}_k^t$  over a finite set containing  $\theta_k^{\text{cur}}$ ; the minimum over a set is no larger than any member's value.  $\square$

**Theorem C.19** (Round-Wise Surrogate Dissipation). *Fix round  $t$  and client  $k \in \mathcal{S}^t$ , holding  $(Z_k^t, S_k^t, \bar{\omega}_k^t)$  fixed during  $E$  local steps. If each step applies Eq. (17), the surrogate sequence is non-increasing:  $\tilde{J}_k^t(\theta_k^{t,s+1}) \leq \tilde{J}_k^t(\theta_k^{t,s})$  for  $s = 0, 1, \dots, E-1$ , where  $\theta_k^{t,0}$  is the round- $t$  initialization and  $\theta_k^{t,E}$  the uploaded model.  $\tilde{J}_k^t(\theta_k^{t,E}) \leq \tilde{J}_k^t(\theta_k^{t,0})$ , and by Proposition C.10 the exact per-pair objective obeys  $g_k^t(\theta_k^{t,E}) \leq g_k^t(\theta_k^{t,0}) + \sum_{s=0}^{E-1} \rho_k^{t,s}$  with  $\rho_k^{t,s} \geq 0$  via Eq. (48) at  $\theta_k^{t,s}$ .*

*Proof.* Apply Proposition C.18 at each local step with  $\theta_k^{\text{cur}} = \theta_k^{t,s}$  and  $\theta_k^+ = \theta_k^{t,s+1}$ ; telescoping over  $s$  gives the round inequality. The per-pair consequence chains Proposition C.10 over the  $E$  steps, so the residuals add.  $\square$

## D. Appendix for the Theoretical Analysis

This appendix proves the convergence results of Section 4, covering both the idealized reference solver and its federated realization. The analysis includes stationarity of simultaneous descent, the surrogate error budget  $\delta_k^t$ , the  $\mathcal{O}(1/\sqrt{T})$  convergence rate, and the lift to the global potential  $V$ .

**Notation.**  $\mathcal{W} = \prod_k \Theta_k$  carries the product metric  $g = \oplus_k g_k$ ;  $\text{grad } V$  is the Riemannian gradient on  $(\mathcal{W}, g)$  and  $\text{grad}_{g_k, \theta_k} V$  its  $k$ -th block. At round  $t$ ,  $\mathcal{S}^t$  is the participating set,  $\mathcal{N}_k^t = \mathcal{N}_k \cap \mathcal{S}^t$ ,  $\mathcal{S}_k^t = \bigcup_{j \in \mathcal{N}_k^t} \mathcal{S}_{kj} \subseteq \mathcal{M}_k$ ,  $\bar{\omega}_k^t = \sum_{j \in \mathcal{N}_k^t} \omega_{kj}$ , and  $z_j^t = \mathcal{T}_j(\theta_j^t)$ . The coordinatewise quantities are:  $\Omega_m^t = \sum_{j \in \mathcal{N}_k^t(m)} \omega_{kj}$ ,  $\nu_m^t = \sum_{j \in \mathcal{N}_k^t(m)} \omega_{kj} \|z_j^{t, (m)} - Z_k^{t, (m)}\|^2$ , and  $C_k^t = \frac{\lambda}{2} \sum_{m \in \mathcal{S}_k^t} \nu_m^t$ .

### D.1. Standing Assumptions

We work under Assumptions 4.1–4.3 (geometry/regularity,  $L$ -smoothness *w.r.t.* retraction of both  $V$  and every round- $t$  surrogate  $\tilde{J}_k^t$ , and lower boundedness with the optional compact sublevel set), and—for the stochastic results—Assumptions 4.8 (unbiased, variance- $\sigma^2$  gradients) and 4.9 (a bounded inter-round surrogate-drift budget  $\Delta_T$ ). We also adopt the coordinatewise masked-Euclidean structure of Eq. (41), under which  $\pi_S \circ \pi_{S'} = \pi_S$  for  $S \subseteq S'$ .

**Second-Order Retraction.** We take the  $L$ -smoothness inequalities of Assumption 4.2 as primitive. A standard sufficient condition is that the relevant iterates lie in a compact region on which the pullbacks through the second-order retraction have uniformly bounded Riemannian Hessians. Assumption 4.3(ii) provides such a region for the reference solver; for GeoEvo, the same uniform bound is assumed along the visited surrogate iterates.

### D.2. Exact Potential and Stationarity of the Reference Solver

**Exact Potential.** Proposition 4.4 restates that  $V$  of Eq. (3) is an exact potential for the game Eq. (2); its proof is given in Proposition B.11. We use only its corollary, the blockwise gradient alignment  $\text{grad}_{g_k, \theta_k} J_k(\theta_k; W_{-k}) = \text{grad}_{g_k, \theta_k} V(W)$  (Lemma B.14), which lets us analyze  $V$  directly.

**Per-Step Decrease.** The reference solver is the simultaneous retracted update of Eq. (9): at each iteration *all* blocks move along their own Riemannian gradient,

$$W^{t+1} = \left( \text{Retr}_{\theta_k^t}(-\eta_t \text{grad}_{g_k, \theta_k} V(W^t)) \right)_{k=1}^K, \quad (49)$$

which is the single product-manifold step  $W^{t+1} = \text{Retr}_{W^t}(-\eta_t \text{grad } V(W^t))$ . Lemma C.3 gives, for  $0 < \eta_t \leq 1/L$ ,

$$V(W^{t+1}) \leq V(W^t) - \frac{\eta_t}{2} \|\text{grad } V(W^t)\|_g^2 = V(W^t) - \frac{\eta_t}{2} \sum_{k=1}^K \|\text{grad}_{g_k, \theta_k} V(W^t)\|_{g_k}^2. \quad (50)$$

We stress that Eq. (50) is obtained from the *joint* step (49), in which no block is held fixed; this is the all-block form required for consistency with the main text.

**Theorem D.1** (Stationarity Convergence). *Under Assumptions 4.2–4.3, the iterates of the simultaneous update (49) with  $0 < \eta_t \leq 1/L$  satisfy: (i)  $\{V(W^t)\}$  is non-increasing; (ii)  $\sum_{t=0}^\infty \eta_t \|\text{grad } V(W^t)\|_g^2 \leq 2(V(W^0) - V_{\inf}) < \infty$ ; (iii) if  $\inf_t \eta_t > 0$  then  $\|\text{grad } V(W^t)\|_g \rightarrow 0$ , hence  $\|\text{grad}_{g_k, \theta_k} V(W^t)\|_{g_k} \rightarrow 0$  for every  $k$  simultaneously; (iv) under the compact sublevel set (Assumption 4.3(ii)) and continuity of  $\text{grad } V$  (ensured by Assumption 4.2), every limit point of  $\{W^t\}$  is a first-order Nash equilibrium.*

*Proof.* (i) is immediate from Eq. (50), whose right side is  $\leq V(W^t)$ . (ii): summing Eq. (50) over  $t = 0, \dots, T-1$  telescopes the left side,  $\sum_{t=0}^{T-1} \frac{\eta_t}{2} \|\text{grad } V(W^t)\|_g^2 \leq V(W^0) - V(W^T) \leq V(W^0) - V_{\inf}$ , using  $V \geq V_{\inf}$  (Assumption 4.3(i)); let  $T \rightarrow \infty$ . (iii): if  $\inf_t \eta_t = \underline{\eta} > 0$ , then  $\underline{\eta} \sum_t \|\text{grad } V(W^t)\|_g^2 \leq 2(V(W^0) - V_{\inf}) < \infty$ , so the summand  $\rightarrow 0$ ; the block identity  $\|\text{grad } V\|_g^2 = \sum_k \|\text{grad}_{g_k, \theta_k} V\|_{g_k}^2$  forces *every* block gradient to vanish in the limit. (iv): by (i) the iterates remain in the compact sublevel set, so a convergent subsequence  $W^{t_\ell} \rightarrow W^\infty$  exists; continuity of  $\text{grad } V$  and (iii) give  $\text{grad } V(W^\infty) = 0$ , i.e.  $W^\infty$  is a first-order Nash equilibrium (Definition B.12, via Lemma B.14).  $\square$

*Remark D.2* (Reference Solver vs. GeoEvo). Theorem D.1 describes the idealized solver that evaluates the exact grad  $V$  and therefore needs per-step neighbor synchronization. GeoEvo replaces this exact coupling by stale broadcast prototypes under partial participation; the relevant federated guarantees are the surrogate-stationarity rate (Theorem D.7) and the lift to  $V$  (Proposition D.10).

### D.3. Optimal Surrogate Approximation and the Error-Bound Lemma

Recall the exact per-pair coupling and the single-prototype surrogate coupling,

$$\Phi_k^t(z) \triangleq \frac{\lambda}{2} \sum_{j \in \mathcal{N}_k^t} \omega_{kj} d_{\mathcal{Z}}^2(\pi_{S_{kj}}(z), \pi_{S_{kj}}(z_j^t)), \quad (51)$$

$$\tilde{\Phi}_k^t(z) \triangleq \frac{\lambda}{2} \bar{\omega}_k^t d_{\mathcal{Z}}^2(\pi_{S_k^t}(z), \pi_{S_k^t}(Z_k^t)), \quad (52)$$

so that the client surrogate is  $\tilde{J}_k^t(\theta_k) = w_k F_k(\theta_k; \mathcal{I}_k) + \tilde{\Phi}_k^t(\mathcal{T}_k(\theta_k))$  (Eq. (12)). We write  $\Phi_k^t$  for what was informally called  $\Phi_k^{t, \text{true}}$ : it is the genuine per-pair object, *not* the union-mask proxy.

**Proposition D.3** (Variational Optimality of the Prototype). *The Fréchet prototype  $Z_k^t$  of Eq. (11) minimizes the exact per-pair coupling:  $\Phi_k^t(Z_k^t) \leq \Phi_k^t(z)$  for all  $z \in \mathcal{Z}$ , with  $\Phi_k^t(Z_k^t) = C_k^t$ . Coordinatewise,  $Z_k^{t, (m)}$  is the weighted Fréchet mean of the neighbors possessing  $m$ .*

*Proof.* By the bias–variance decomposition Lemma C.8,  $\Phi_k^t(z) = \frac{\lambda}{2} \sum_{m \in S_k^t} \Omega_m^t \|z^{(m)} - Z_k^{t, (m)}\|^2 + C_k^t$ . The first non-negative term is minimized, and equals 0, exactly at  $z^{(m)} = Z_k^{t, (m)}$  for every  $m \in S_k^t$ ; hence  $Z_k^t$  minimizes  $\Phi_k^t$  with value  $C_k^t$ . The coordinatewise weighted-mean form is Eq. (43).  $\square$

We now derive an explicit two-sided bound on the round- $t$  iterates and candidates evaluated by GeoEvo, which is sufficient for all uses of Assumption 4.7 in the subsequent analysis.

**Lemma D.4** (Surrogate Error Bound). *If  $\mathcal{N}_k^t = \emptyset$ , the claim holds with  $\delta_k^t = 0$ . Hence assume  $\mathcal{N}_k^t \neq \emptyset$ . Let  $\mathcal{K}_k^t \subseteq \Theta_k$  contain all client iterates and candidates evaluated in round  $t$ , and assume that their structural representations are bounded. Define*

$$R_k^t \triangleq \sup_{\theta_k \in \mathcal{K}_k^t} d_{\mathcal{Z}}(\pi_{S_k^t}(\mathcal{T}_k(\theta_k)), \pi_{S_k^t}(Z_k^t)) < \infty,$$

and

$$\Omega_k^{t, \min} \triangleq \min_{m \in S_k^t} \Omega_m^t.$$

Then, for every  $\theta_k \in \mathcal{K}_k^t$ ,

$$\left| \Phi_k^t(\mathcal{T}_k(\theta_k)) - \tilde{\Phi}_k^t(\mathcal{T}_k(\theta_k)) \right| \leq \delta_k^t, \quad (53)$$

where

$$\delta_k^t \triangleq \underbrace{C_k^t}_{\text{dispersion}} + \underbrace{\frac{\lambda}{2} (\bar{\omega}_k^t - \Omega_k^{t, \min}) (R_k^t)^2}_{\text{per-pair-to-union-mask collapse}}.$$

The mask-collapse term vanishes when all positive-weight neighbors have the same overlap support  $S_k^t$ , while the dispersion term vanishes when the participating neighbors agree on every shared coordinate.

*Proof.* Let  $z = \mathcal{T}_k(\theta_k)$ , and  $D_m \triangleq \|z^{(m)} - Z_k^{t, (m)}\|^2$ . By the definition of  $R_k^t$ ,

$$\sum_{m \in S_k^t} D_m = d_{\mathcal{Z}}^2(\pi_{S_k^t}(z), \pi_{S_k^t}(Z_k^t)) \leq (R_k^t)^2.$$

Lemma C.8 and Eq. (52) give

$$\Phi_k^t(z) - \tilde{\Phi}_k^t(z) = C_k^t - \frac{\lambda}{2} \sum_{m \in S_k^t} (\bar{\omega}_k^t - \Omega_m^t) D_m.$$

Since  $0 \leq \Omega_m^t \leq \bar{\omega}_k^t$ ,

$$0 \leq \sum_{m \in S_k^t} (\bar{\omega}_k^t - \Omega_m^t) D_m \leq (\bar{\omega}_k^t - \Omega_k^{t,\min}) (R_k^t)^2.$$

Therefore,

$$\left| \Phi_k^t(z) - \tilde{\Phi}_k^t(z) \right| \leq C_k^t + \frac{\lambda}{2} (\bar{\omega}_k^t - \Omega_k^{t,\min}) (R_k^t)^2 = \delta_k^t.$$

□

**Remark D.5** (Explicit Decomposition of the Surrogate Error  $\delta_k^t$ ). Earlier drafts bounded  $\tilde{\Phi}_k^t$  against a union-mask object, capturing only neighbor compression ( $C_k^t$ ) and silently ignoring the change of mask from the edge-specific  $\pi_{S_{kj}}$  to the union  $\pi_{S_k^t}$ . Lemma D.4 compares  $\tilde{\Phi}_k^t$  against the genuine per-pair coupling  $\Phi_k^t$  (Eq. (51)) and isolates the missing geometry as the mask-collapse term  $\frac{\lambda}{2} (\bar{\omega}_k^t - \Omega_k^{t,\min}) (R_k^t)^2$ , which is non-negative by the masked-distance monotonicity (Lemma C.5) and the weight ordering  $\Omega_m^t \leq \bar{\omega}_k^t$ . Both sources are explicit and admit clean vanishing regimes; hence, on the evaluated round- $t$  iterates and candidates,  $\delta_k^t$  is explicit rather than opaque.

#### D.4. Stochastic Convergence to Surrogate Stationarity

We use the participating-average surrogate  $\tilde{J}^t \triangleq |\mathcal{S}^t|^{-1} \sum_{k \in \mathcal{S}^t} \tilde{J}_k^t(\theta_k^t)$ , let  $\tilde{J}_{\inf} > -\infty$  be a common lower bound on the corresponding round-input and round-output averages, and write  $\langle X_k \rangle_t \triangleq |\mathcal{S}^t|^{-1} \sum_{k \in \mathcal{S}^t} X_k$ . For local-step indexing, set  $\theta_k^{t,0} \triangleq \theta_k^t$  and  $\theta_k^{t,E} \triangleq \theta_k^{t+1}$ .

**Lemma D.6** (Filtered Riemannian-SGD Descent of the Surrogate). *Fix round  $t$ , client  $k \in \mathcal{S}^t$ , and the message  $(Z_k^t, S_k^t, \bar{\omega}_k^t)$ . Let  $\theta_k^{\text{base},t,0}$  be the base candidate computed at  $\theta_k^{t,0}$  using an unbiased stochastic gradient estimator and stepsize  $\eta \leq 1/L$ . Let  $\theta_k^{t,1}$  be the first accepted iterate and  $\theta_k^{t,E} = \theta_k^{t+1}$  the final output after  $E$  accepted local updates. Then*

$$\mathbb{E} \left[ \tilde{J}_k^t(\theta_k^{t+1}) \mid \theta_k^t \right] \leq \tilde{J}_k^t(\theta_k^t) - \frac{\eta}{2} \left\| \text{grad}_{g_k, \theta_k^t} \tilde{J}_k^t \right\|_{g_k}^2 + \frac{L\eta^2}{2} \sigma^2. \quad (54)$$

*Proof.* Conditioning on the fixed round- $t$  message, the  $L$ -smoothness of  $\tilde{J}_k^t$  and the definition  $\theta_k^{\text{base},t,0} = \text{Retr}_{\theta_k^{t,0}}(-\eta \widehat{\text{grad}}_{g_k, \theta_k^{t,0}} \tilde{J}_k^t)$  give

$$\tilde{J}_k^t(\theta_k^{\text{base},t,0}) \leq \tilde{J}_k^t(\theta_k^{t,0}) - \eta \left\langle \text{grad}_{g_k, \theta_k^{t,0}} \tilde{J}_k^t, \widehat{\text{grad}}_{g_k, \theta_k^{t,0}} \tilde{J}_k^t \right\rangle_{g_k} + \frac{L\eta^2}{2} \left\| \widehat{\text{grad}}_{g_k, \theta_k^{t,0}} \tilde{J}_k^t \right\|_{g_k}^2.$$

By Assumption 4.8,

$$\mathbb{E} \left[ \left\| \widehat{\text{grad}}_{g_k, \theta_k^{t,0}} \tilde{J}_k^t \right\|_{g_k}^2 \mid \theta_k^{t,0} \right] \leq \left\| \text{grad}_{g_k, \theta_k^{t,0}} \tilde{J}_k^t \right\|_{g_k}^2 + \sigma^2.$$

Taking conditional expectations and using  $\eta \leq 1/L$  therefore yield

$$\mathbb{E} \left[ \tilde{J}_k^t(\theta_k^{\text{base},t,0}) \mid \theta_k^{t,0} \right] \leq \tilde{J}_k^t(\theta_k^{t,0}) - \frac{\eta}{2} \left\| \text{grad}_{g_k, \theta_k^{t,0}} \tilde{J}_k^t \right\|_{g_k}^2 + \frac{L\eta^2}{2} \sigma^2.$$

At the first local update, the candidate pool contains  $\theta_k^{\text{base},t,0}$ . Each subsequent selection retains its current iterate, and the server field remains fixed throughout round  $t$ . Consequently, pointwise for every realization,

$$\tilde{J}_k^t(\theta_k^{t+1}) = \tilde{J}_k^t(\theta_k^{t,E}) \leq \tilde{J}_k^t(\theta_k^{t,1}) \leq \tilde{J}_k^t(\theta_k^{\text{base},t,0}).$$

Taking conditional expectations proves Eq. (54). □

**Theorem D.7** (Non-Convex Rate to Surrogate Stationarity). *Under Assumptions 4.2, 4.8, 4.9, the GeoEvo iterates with  $\eta \leq 1/L$  satisfy, with  $C = 2$ ,*

$$\min_{t < T} \mathbb{E} \left[ \left\langle \left\| \text{grad}_{g_k} \tilde{J}_k^t(\theta_k^t) \right\|_{g_k}^2 \right\rangle_t \right] \leq \frac{C(\mathbb{E}[\tilde{J}^0] - \tilde{J}_{\inf} + \Delta_T)}{\eta T} + CL\eta\sigma^2. \quad (55)$$

*Proof.* Average Eq. (54) over  $k \in \mathcal{S}^t$  and rearrange:

$$\frac{\eta}{2} \mathbb{E} \langle \|\text{grad}_{g_k} \tilde{J}_k^t(\theta_k^t)\|_{g_k}^2 \rangle_t \leq \mathbb{E} \langle \tilde{J}_k^t(\theta_k^t) \rangle_t - \mathbb{E} \langle \tilde{J}_k^t(\theta_k^{t+1}) \rangle_t + \frac{L\eta^2}{2} \sigma^2. \quad (56)$$

The first term is  $\mathbb{E}[\tilde{J}^t]$ . The drift budget  $\Delta_T$  includes changes in the participating set, server prototype, union support, and aggregate coupling weight. For  $t = 0, \dots, T-2$ , the definition of  $\xi_t$  in Assumption 4.9 gives

$$\frac{1}{|\mathcal{S}^t|} \sum_{k \in \mathcal{S}^t} \tilde{J}_k^t(\theta_k^{t+1}) = \tilde{J}^{t+1} - \xi_t \geq \tilde{J}^{t+1} - [\xi_t]_+.$$

Substituting this inequality into Eq. (56), summing over  $t = 0, \dots, T-1$ , and using the common lower bound  $\tilde{J}_{\inf}$  for the final round-output surrogate yield

$$\begin{aligned} \frac{\eta}{2} \sum_{t=0}^{T-1} \mathbb{E} \left[ \left\langle \left\| \text{grad}_{g_k} \tilde{J}_k^t(\theta_k^t) \right\|_{g_k}^2 \right\rangle_t \right] &\leq \mathbb{E}[\tilde{J}^0] - \tilde{J}_{\inf} + \sum_{t=0}^{T-2} \mathbb{E}[[\xi_t]_+] + \frac{TL\eta^2}{2} \sigma^2 \\ &\leq \mathbb{E}[\tilde{J}^0] - \tilde{J}_{\inf} + \Delta_T + \frac{TL\eta^2}{2} \sigma^2. \end{aligned}$$

Dividing by  $\frac{\eta}{2}T$  and bounding the average from below by the minimum over  $t < T$  gives Eq. (55) with  $C = 2$ .  $\square$

**Corollary D.8** (Step-Size Schedule). *With  $\eta = \Theta(1/\sqrt{T})$ , the third term of Eq. (55) is  $\mathcal{O}(1/\sqrt{T})$  and the first is  $\mathcal{O}((1 + \Delta_T)/\sqrt{T})$ . Hence: (i) if the drift is summable,  $\Delta_T = \mathcal{O}(1)$ , the rate is  $\mathcal{O}(1/\sqrt{T})$ ; (ii) if  $\Delta_T = o(\sqrt{T})$ , the bound is  $o(1)$  (asymptotic surrogate stationarity); (iii) if  $\Delta_T = \Theta(\sqrt{T})$ , the bound saturates at an  $\mathcal{O}(1)$  stationarity neighborhood. Regime (i) is the clean  $\mathcal{O}(1/\sqrt{T})$  statement of Theorem 4.10; the others make explicit that vanishing stationarity requires the inter-round surrogate drift to be controlled.*

The selection filter is thus a *safe-exploration* mechanism: by Lemma D.6 it never degrades the worst-case Riemannian-SGD bound, while the NES/PSO candidates provide nonlocal exploratory proposals.

### D.5. Lifting to the Exact Potential: Expected Dissipation of $V$

We now lift the surrogate descent guarantees to the global exact potential  $V$ . The appendix C provides the per-round, per-client structural half via the majorization residual  $\rho_k^t$  (Proposition C.10). Here we incorporate three additional real-world factors: partial participation ( $\mathcal{S}^t$ ), inter-round variation of the surrogate field (bounded by  $\Delta_T$ ), and stochastic gradient noise. The result is an expected dissipation inequality for  $V$  and the conditional exact-potential stationarity bound of Proposition D.10.

Let  $g_k^t(\theta) \triangleq w_k F_k(\theta; \mathcal{I}_k) + \Phi_k^t(\mathcal{T}_k(\theta))$  be the exact per-pair block over the participating neighbors  $\mathcal{N}_k^t$ . At the round- $t$  initialization, these neighbors sit exactly at their uploads  $z_j^t$ , so  $g_k^t$  coincides with the participating part of the  $\theta_k$ -block of  $V$  at  $W^t$ . Under full participation ( $\mathcal{N}_k^t = \mathcal{N}_k$ ) this is the entire block and  $\text{grad}_{g_k} g_k^t(\theta_k^t) = \text{grad}_{g_k, \theta_k} V(W^t)$ ; under partial participation the omitted edges  $\{(k, j) : j \in \mathcal{N}_k \setminus \mathcal{N}_k^t\}$  contribute a further bounded term  $\beta_k^t \triangleq \|\text{grad}_{g_k, \theta_k} V(W^t) - \text{grad}_{g_k} g_k^t(\theta_k^t)\|_{g_k}^2$ , which we carry explicitly in the residual below. Here  $\beta_k^t = 0$  under full participation and can be bounded in terms of the omitted neighbor mass when the edge gradients are uniformly bounded.

**Assumption D.9** (Controlled Federated-Lift Residual). Let  $\chi_t \triangleq V(W^{t+1}) - V(W^t) - \sum_{k \in \mathcal{S}^t} [g_k^t(\theta_k^{t+1}) - g_k^t(\theta_k^t)]$  denote the residual between the simultaneous change of the exact potential and the sum of the participating exact-block changes. We assume  $\sum_{t=0}^{T-1} \mathbb{E}[[\chi_t]_+] \leq \Xi_T$ ,  $[a]_+ \triangleq \max\{a, 0\}$ .

**Proposition D.10** (Conditional Lift to the Exact Potential). *Under the assumptions of Theorem D.7, Assumption D.9, and Lemma D.4, suppose additionally that, for some  $\kappa > 0$  and every  $t < T$ ,  $k \in \mathcal{S}^t$ ,*

$$\left\| \text{grad}_{g_k} \tilde{J}_k^t(\theta_k^t) - \text{grad}_{g_k} g_k^t(\theta_k^t) \right\|_{g_k}^2 \leq \kappa \delta_k^t. \quad (57)$$

Then, for each  $k \in \mathcal{S}^t$ ,

$$\mathbb{E}[g_k^t(\theta_k^{t+1}) \mid \theta_k^t] \leq g_k^t(\theta_k^t) - \frac{\eta}{2} \left\| \text{grad}_{g_k} \tilde{J}_k^t(\theta_k^t) \right\|_{g_k}^2 + \frac{L\eta^2}{2} \sigma^2 + 2\delta_k^t. \quad (58)$$

Moreover,

$$\mathbb{E}[V(W^{t+1}) - V(W^t)] \leq -\frac{\eta}{2} \sum_{k \in \mathcal{S}^t} \mathbb{E} \left[ \left\| \text{grad}_{g_k} \tilde{J}_k^t(\theta_k^t) \right\|_{g_k}^2 \right] + \frac{|\mathcal{S}^t| L \eta^2}{2} \sigma^2 + 2 \sum_{k \in \mathcal{S}^t} \delta_k^t + \mathbb{E}[\chi_t]_+. \quad (59)$$

Furthermore,

$$\left\| \text{grad}_{g_k, \theta_k} V(W^t) \right\|_{g_k}^2 \leq 3 \left\| \text{grad}_{g_k} \tilde{J}_k^t(\theta_k^t) \right\|_{g_k}^2 + 3\kappa \delta_k^t + 3\beta_k^t. \quad (60)$$

Consequently,

$$\min_{t \leq T} \mathbb{E} \left[ \left\langle \left\| \text{grad}_{g_k, \theta_k} V(W^t) \right\|_{g_k}^2 \right\rangle_t \right] \leq \frac{3C(\mathbb{E}[\tilde{J}^0] - \tilde{J}_{\inf} + \Delta_T)}{\eta T} + 3CL\eta\sigma^2 + 3\kappa\bar{\delta}_T + 3\bar{\beta}_T, \quad (61)$$

where

$$\bar{\delta}_T \triangleq \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\langle \delta_k^t \rangle_t], \quad \bar{\beta}_T \triangleq \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\langle \beta_k^t \rangle_t]. \quad (62)$$

Thus, if  $\eta = \Theta(T^{-1/2})$ ,  $\Delta_T = \mathcal{O}(1)$ ,  $\bar{\delta}_T \rightarrow 0$ , and  $\bar{\beta}_T \rightarrow 0$ , the iterates approach first-order stationarity of  $V$ . The cumulative expected-dissipation residual in Eq. (59) is controlled separately by  $\Xi_T$ .

**Proof. Per-Round Block Inequality.** By Lemma D.4,  $g_k^t(\theta) \leq \tilde{J}_k^t(\theta) + \delta_k^t$ ,  $\tilde{J}_k^t(\theta) \leq g_k^t(\theta) + \delta_k^t$  on the round- $t$  iterates and candidates. Combining these inequalities with Eq. (54) gives

$$\begin{aligned} \mathbb{E}[g_k^t(\theta_k^{t+1}) \mid \theta_k^t] &\leq \mathbb{E}[\tilde{J}_k^t(\theta_k^{t+1}) \mid \theta_k^t] + \delta_k^t \\ &\leq \tilde{J}_k^t(\theta_k^t) - \frac{\eta}{2} \left\| \text{grad}_{g_k} \tilde{J}_k^t(\theta_k^t) \right\|_{g_k}^2 + \frac{L\eta^2}{2} \sigma^2 + \delta_k^t \\ &\leq g_k^t(\theta_k^t) - \frac{\eta}{2} \left\| \text{grad}_{g_k} \tilde{J}_k^t(\theta_k^t) \right\|_{g_k}^2 + \frac{L\eta^2}{2} \sigma^2 + 2\delta_k^t, \end{aligned}$$

which proves Eq. (58).

**Lift to the Exact Potential.** By the definition of  $\chi_t$  in Assumption D.9,

$$\begin{aligned} V(W^{t+1}) - V(W^t) &= \sum_{k \in \mathcal{S}^t} [g_k^t(\theta_k^{t+1}) - g_k^t(\theta_k^t)] + \chi_t \\ &\leq \sum_{k \in \mathcal{S}^t} [g_k^t(\theta_k^{t+1}) - g_k^t(\theta_k^t)] + [\chi_t]_+. \end{aligned}$$

Taking expectations and applying Eq. (58) proves Eq. (59). Summing it over  $t = 0, \dots, T-1$  and using  $\sum_{t=0}^{T-1} \mathbb{E}[\chi_t]_+ \leq \Xi_T$  shows that the cumulative lift residual is controlled by  $\Xi_T$ .

**Stationarity of the Exact Potential.** At each round,

$$\text{grad}_{g_k, \theta_k} V(W^t) = \text{grad}_{g_k} \tilde{J}_k^t(\theta_k^t) + \left[ \text{grad}_{g_k} g_k^t(\theta_k^t) - \text{grad}_{g_k} \tilde{J}_k^t(\theta_k^t) \right] + \left[ \text{grad}_{g_k, \theta_k} V(W^t) - \text{grad}_{g_k} g_k^t(\theta_k^t) \right].$$

Using  $\|a + b + c\|^2 \leq 3(\|a\|^2 + \|b\|^2 + \|c\|^2)$ , Eq. (57), and the definition of  $\beta_k^t$  proves Eq. (60).

Averaging Eq. (60) over  $k \in \mathcal{S}^t$  and then over  $t = 0, \dots, T-1$  yields

$$\min_{t \leq T} \mathbb{E} \left[ \left\langle \left\| \text{grad}_{g_k, \theta_k} V(W^t) \right\|_{g_k}^2 \right\rangle_t \right] \leq \frac{3}{T} \sum_{t=0}^{T-1} \mathbb{E} \left[ \left\langle \left\| \text{grad}_{g_k} \tilde{J}_k^t(\theta_k^t) \right\|_{g_k}^2 \right\rangle_t \right] + 3\kappa\bar{\delta}_T + 3\bar{\beta}_T.$$

The averaged descent inequality established in the proof of Theorem D.7 bounds the first term by  $\frac{3C(\mathbb{E}[\tilde{J}^0] - \tilde{J}_{\inf} + \Delta_T)}{\eta T} + 3CL\eta\sigma^2$ . Substitution gives Eq. (61).  $\square$

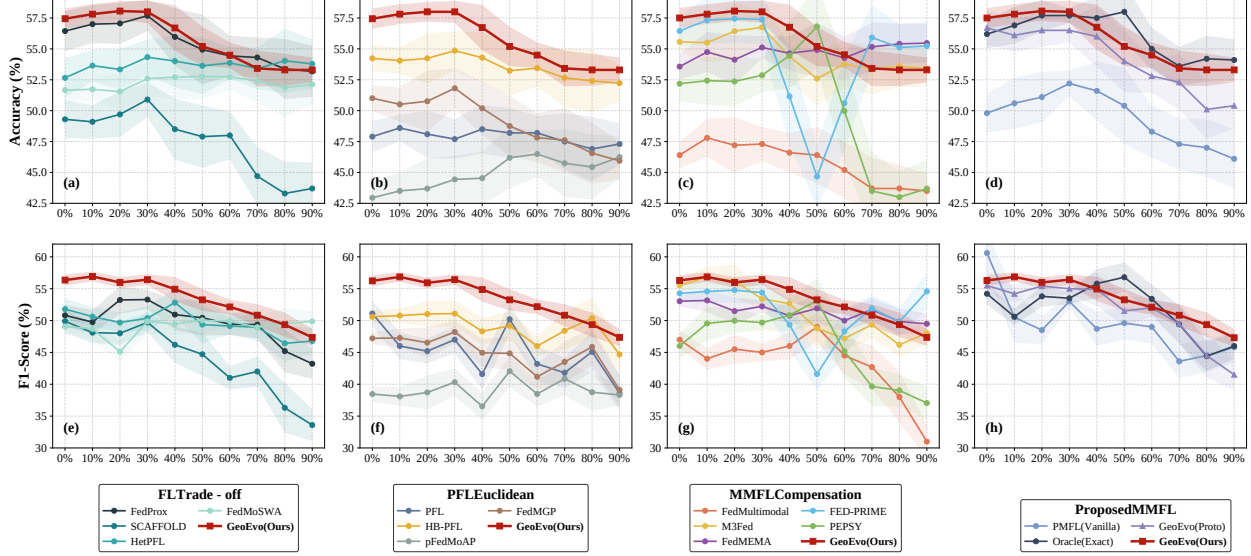


Figure 3. Comparative performance of representative methods on the UPMC Food-101 dataset under binomial modality simulation.

## D.6. Monotone Surrogate Descent of Evolutionary Proposals

**Proposition D.11** (Evolutionary Proposals Preserve Monotonicity). *Fix round  $t$  and fields  $\{Z_k^t\}$ , and let  $\theta_k^t$  be any proposal (gradient-based, NES, or PSO). If the client applies the selection rule in Eq. (17), which retains  $\theta_k^{\text{cur}}$  in the pool, then*

$$\tilde{J}_k^t(\theta_k^{t+1}) \leq \tilde{J}_k^t(\theta_k^t).$$

By Lemma D.4, the exact per-pair coupling contribution obeys

$$g_k^t(\theta_k^{t+1}) \leq g_k^t(\theta_k^t) + 2\delta_k^t.$$

*Proof.* At local step  $s$ , the candidate pool contains the current iterate  $\theta_k^{t,s}$ . Hence the finite-pool minimization in Eq. (17) gives

$$\tilde{J}_k^t(\theta_k^{t,s+1}) \leq \tilde{J}_k^t(\theta_k^{t,s}), \quad s = 0, \dots, E-1,$$

independently of how the gradient, NES, or PSO proposals are generated. Iterating this inequality over the  $E$  local updates yields

$$\tilde{J}_k^t(\theta_k^{t+1}) = \tilde{J}_k^t(\theta_k^{t,E}) \leq \tilde{J}_k^t(\theta_k^{t,0}) = \tilde{J}_k^t(\theta_k^t).$$

For the exact coupling, the two-sided bound of Lemma D.4 applies at both the round input and output. Therefore,

$$g_k^t(\theta_k^{t+1}) \leq \tilde{J}_k^t(\theta_k^{t+1}) + \delta_k^t \leq \tilde{J}_k^t(\theta_k^t) + \delta_k^t \leq g_k^t(\theta_k^t) + 2\delta_k^t.$$

The factor 2 arises because the approximation bound is applied once at each endpoint.  $\square$

**Remark D.12** (Saddle Avoidance in Practice). The stochastic exploratory nature of the NES/PSO proposals, combined with the monotone filter, may help GeoEvo discover descent directions in regions of negative curvature where a first-order step stalls. Because candidates are selected according to  $\tilde{J}_k^t$ , this exploration remains surrogate-controlled and preserves the per-round guarantees above.

**Remark D.13** (Exploration at No Cost to the Rate). Because the base candidate computed from an unbiased stochastic gradient estimator remains in the candidate pool, accepting any NES/PSO proposal cannot increase  $\tilde{J}_k^t$ . Consequently, the worst-case surrogate-stationarity rate of Theorem D.7 is exactly preserved, and a successful exploratory move strictly improves the per-step objective. The sole slack incurred when lifting this guarantee to the exact per-pair objective  $g_k^t$  is the additive term  $2\delta_k^t$ , which is controlled by Lemma D.4 and vanishes under homogeneous overlaps and vanishing neighbor dispersion. Hence, evolutionary exploration is *safe by construction*: it can propose moves beyond local basins yet never degrade the monotone descent or the convergence rate established above.

## E. Experiments

### E.1. Experimental Setting

#### E.1.1. DATASETS

We evaluate our method across diverse tasks spanning multiple domains. For emotion recognition (ER), we use CREMA-D (Cao et al., 2014), which contains audio-visual clips from 91 actors performing 12 sentences in six emotions (neutral, happy, anger, disgust, fear, sad). For social media (SM) crisis classification, we use CrisisMMD (Alam et al., 2018), a dataset of tweets with paired image and text data from seven major natural disasters, annotated for disaster impact such as property damage, injuries, and fatalities. For large-scale classification, we use UPMC Food-101 (Wang et al., 2015), a noisy bimodal dataset of image-text pairs from 101 food categories, serving as a benchmark for robust alignment under high-dimensional modality incompleteness.

#### E.1.2. EVALUATION METRICS

To ensure a rigorous and context-sensitive assessment, we adopt a tailored evaluation strategy where the choice of metrics aligns with the specific characteristics of each multimodal learning task. For the emotion recognition task, where class distribution is often imbalanced, we employ Accuracy (ACC), Unweighted Average Recall (UAR), and the F1 score. These metrics collectively quantify overall correctness, per-class sensitivity, and the harmonic balance between precision and recall, respectively. Social media crisis classification and food category recognition (UPMC Food-101) are evaluated using Accuracy and the F1 score, providing a standard measure of classification effectiveness in these contexts.

#### E.1.3. PARTITION AND SIMULATION

To simulate realistic federated conditions, we construct non-IID client partitions and missing-modality patterns tailored to each dataset. Speaker-dependent emotion data (CREMA-D) is partitioned by speaker ID, capturing natural individual variations. For CrisisMMD and UPMC Food-101, we generate synthetic non-IID splits using a Dirichlet distribution  $\text{Dir}(\alpha)$  with  $\alpha \in \{0.05, 0.1, 0.5, 1.0\}$  to control heterogeneity levels, where smaller values of  $\alpha$  induce more severe distribution skew while larger values yield more uniform distributions.

We implement two client-level static-missingness protocols. *Proportional simulation* follows the dataset-specific distribution of client modality availability. *Binomial simulation* draws one modality mask per client from modality-wise Bernoulli probabilities at initialization and keeps it fixed throughout training and evaluation.

#### E.1.4. IMPLEMENTATION DETAILS

Following FedMultimodal (Feng et al., 2023), we employ efficient pre-trained backbone models for feature extraction: MobileNetV2 (Howard et al., 2017) for vision, MobileBERT (Sun et al., 2020) for text, and MFCCs (Davis & Mermelstein, 1980) for audio. For the large-scale UPMC Food-101, we adopt the ViLT (Kim et al., 2021) encoder to handle high-dimensional alignment. Our GeoEvo employs a lightweight multimodal architecture consisting of: (i) modality-specific encoders (RNN-only for video and text, Conv-RNN for others); (ii) an attention-based fusion module; and (iii) a two-layer dense classifier. The fusion module is implemented as a single-layer structured self-attention mechanism. For CREMA-D, we use 4 attention heads with hidden dimensions of 64 and 128, respectively. For CrisisMMD, we employ a fused-attention variant that operates on concatenated features, configured with 6 heads and a hidden dimension of 64 to capture structured cross-modal interactions better. All experiments share the following fixed hyperparameters: batch size of 16, dropout rate of 0.2, ReLU activation, and the AdamW optimizer with linear learning rate decay. These consistent settings ensure fair and reproducible comparisons across all evaluations.

### E.2. Cross-Dataset Robustness Analysis

Figures 2 and 3 show GeoEvo maintaining the strongest overall performance envelope as modality missingness and statistical heterogeneity increase. Its advantage generally widens at high missing rates and small Dirichlet parameters, although occasional reversals are consistent with mild missingness reducing shared-model bias or extreme joint heterogeneity sparsifying compatible neighbors and increasing prototype dispersion. The progression from PMFL (Vanilla) through GeoEvo (Prototype) to GeoEvo (Ours) supports the complementary benefits of identity-aware prototype transfer and NES/PSO exploration across both datasets.

Table 2. Notation Summary.

| Category                                        | Symbol                                                                             | Description                                                                                                            |
|-------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Clients, modalities, and data</b>            |                                                                                    |                                                                                                                        |
| Indices                                         | $K, [K], k, j$                                                                     | Number of clients, client index set $[K] \triangleq \{1, \dots, K\}$ , and client indices.                             |
| Rounds and steps                                | $T, t, E, s$                                                                       | Number of rounds, round index, local steps per round, and local-step index.                                            |
| Modalities                                      | $\mathcal{M}, \mathcal{M}_k, m$                                                    | Modality universe, fixed modality set of client $k$ , and modality index.                                              |
| Input spaces                                    | $\mathcal{X}^{(m)}, \mathcal{X}_k$                                                 | Modality input space and client space $\mathcal{X}_k = \prod_{m \in \mathcal{M}_k} \mathcal{X}^{(m)}$ .                |
| Data and identity                               | $\mathcal{D}_k, \mathcal{I}_k$                                                     | Local distribution and client identity $\mathcal{I}_k \triangleq (\mathcal{D}_k, \mathcal{M}_k)$ .                     |
| Marginal law                                    | $\mathcal{D}_k _{\mathcal{M}'}$                                                    | Marginal of the full-modality distribution on $\mathcal{M}' \subseteq \mathcal{M}$ .                                   |
| <b>Objectives and potential game</b>            |                                                                                    |                                                                                                                        |
| Parameters                                      | $\theta_g, \theta_k, \theta_k^*$                                                   | Global parameter, client parameter, and local risk minimizer.                                                          |
| Risk                                            | $F_k(\theta; \mathcal{M}'), F_k^*(\mathcal{M}')$                                   | Population risk under $\mathcal{M}'$ and its minimum value.                                                            |
| Weights                                         | $w_k, \lambda$                                                                     | Client aggregation weight and structural-coupling coefficient.                                                         |
| Strategy profile                                | $W, W_{-k}, \mathcal{W}$                                                           | Joint profile, profile excluding client $k$ , and product manifold $\mathcal{W} = \prod_{k=1}^K \Theta_k$ .            |
| Payoff and potential                            | $J_k(\theta_k; W_{-k}), V(W)$                                                      | Exact client payoff and exact global potential.                                                                        |
| Lower bound and equilibrium                     | $V_{\inf}, \text{FONE}$                                                            | Potential lower bound and first-order Nash equilibrium characterized by $\text{grad } V(W) = 0$ .                      |
| <b>Riemannian and Fisher geometry</b>           |                                                                                    |                                                                                                                        |
| Feasible geometry                               | $\Theta_k, T_{\theta_k} \Theta_k$                                                  | Client parameter manifold and its tangent space at $\theta_k$ .                                                        |
| Metrics                                         | $g_k, g$                                                                           | Client metric and product metric $g = \bigoplus_{k=1}^K g_k$ on $\mathcal{W}$ .                                        |
| Retraction                                      | $\text{Retr}_{\theta_k}, \text{Retr}_W$                                            | Client-level and product-manifold retractions.                                                                         |
| Riemannian gradient                             | $\text{grad}_{g_k, \theta_k}, \text{grad}$                                         | Client-block and product-manifold Riemannian gradients.                                                                |
| Fisher matrices                                 | $\mathbf{F}_k, \bar{\mathbf{F}}_k$                                                 | Local Fisher information and damped matrix $\bar{\mathbf{F}}_k = \mathbf{F}_k + \epsilon I$ .                          |
| Score and population gradient                   | $s_k, g_k$                                                                         | Likelihood score and population-risk gradient; $g_k = -\mathbb{E}[s_k]$ under negative log-likelihood.                 |
| Identifiable subspace                           | $\mathcal{S}_k(\theta)$                                                            | Fisher-identifiable subspace range $\mathbf{F}_k(\theta)$ .                                                            |
| Active coordinates                              | $\mathcal{A}_k, E_{\mathcal{A}_k}, \Pi_{\mathcal{A}_k}$                            | Active indices, their coordinate subspace, and projector.                                                              |
| <b>Transferable structure and compatibility</b> |                                                                                    |                                                                                                                        |
| Structure space                                 | $\mathcal{Z}, \mathcal{Z}^{(m)}, d_{\mathcal{Z}}$                                  | Transferable structure space, modality block, and metric; block-Euclidean in GeoEvo.                                   |
| Structure map                                   | $\mathcal{T}_k, z_k$                                                               | Extraction map $\mathcal{T}_k : \Theta_k \rightarrow \mathcal{Z}$ and representation $z_k = \mathcal{T}_k(\theta_k)$ . |
| Restriction                                     | $\pi_S, \mathcal{Z}_S$                                                             | Restriction to support $S$ and its supported structure space.                                                          |
| Pairwise support                                | $S_{kj}$                                                                           | Shared modality support $S_{kj} = \mathcal{M}_k \cap \mathcal{M}_j$ .                                                  |
| Compatible neighbors                            | $\mathcal{N}_k$                                                                    | Clients $j \neq k$ satisfying $S_{kj} \neq \emptyset$ .                                                                |
| Edge weight                                     | $\omega_{kj}$                                                                      | Symmetric structural-coupling weight on edge $\{k, j\}$ .                                                              |
| Structural dimensions                           | $q_m, q(S),  z $                                                                   | Block, supported, and full structural dimensions.                                                                      |
| <b>Federated prototype construction</b>         |                                                                                    |                                                                                                                        |
| Round participation                             | $\mathcal{S}^t, \mathcal{N}_k^t$                                                   | Round- $t$ clients and compatible subset $\mathcal{N}_k^t = \mathcal{N}_k \cap \mathcal{S}^t$ .                        |
| Union support                                   | $\mathcal{S}_k^t$                                                                  | Union $\bigcup_{j \in \mathcal{N}_k^t} S_{kj}$ of active overlap supports.                                             |
| Aggregate weight                                | $\bar{\omega}_k^t$                                                                 | Aggregate weight $\sum_{j \in \mathcal{N}_k^t} \omega_{kj}$ .                                                          |
| Prototype                                       | $Z_k^t$                                                                            | Server-computed Fréchet prototype broadcast to client $k$ .                                                            |
| Coordinate neighbors                            | $\mathcal{N}_k^t(m), \Omega_m^t$                                                   | Neighbors possessing modality $m$ and their total weight.                                                              |
| Dispersion                                      | $\nu_m^t, C_k^t$                                                                   | Coordinatewise weighted dispersion and $C_k^t = \frac{\lambda}{2} \sum_{m \in \mathcal{S}_k^t} \nu_m^t$ .              |
| <b>GeoEvo surrogate and local search</b>        |                                                                                    |                                                                                                                        |
| Couplings                                       | $\Phi_k^t, \tilde{\Phi}_k^t$                                                       | Exact participating coupling and its single-prototype surrogate.                                                       |
| Client objectives                               | $\tilde{J}_k^t, g_k^t$                                                             | Prototype surrogate and corresponding exact participating block objective.                                             |
| Lift                                            | $\text{Lift}_k(z; \theta_k)$                                                       | Selected minimizer or warm-started approximate structural match to $z$ .                                               |
| Stepsizes and NES scale                         | $\eta, \eta_{\text{NES}}, \sigma_z$                                                | Base and NES stepsizes, and Gaussian smoothing scale.                                                                  |
| NES sampling                                    | $m, \varepsilon_i$                                                                 | NES sample count and perturbations $\varepsilon_i \sim \mathcal{N}(0, I)$ .                                            |
| PSO state                                       | $u_k, z_k^{\text{best}}$                                                           | PSO velocity and client personal-best representation.                                                                  |
| PSO coefficients                                | $c_0, c_1, c_2, r_1, r_2$                                                          | Inertia, attraction coefficients, and uniform random factors.                                                          |
| Candidates                                      | $\theta_k^{\text{base}}, \theta_k^{\text{NES}}, \theta_k^{\text{PSO}}, \theta_k^+$ | Fisher, NES, PSO, and selected candidates.                                                                             |
| Local iterates                                  | $\theta_k^{t,s}, \theta_k^{t,0}, \theta_k^{t,E}$                                   | Local-step iterate, round input, and final round output.                                                               |
| <b>Approximation and convergence analysis</b>   |                                                                                    |                                                                                                                        |
| Smoothness and noise                            | $L, \sigma^2$                                                                      | Retraction-smoothness constant and gradient-noise variance.                                                            |
| Participating average                           | $\tilde{J}^t, \tilde{J}_{\inf}^t, \langle X_k \rangle_t$                           | Participating surrogate average, common lower bound, and average over $\mathcal{S}^t$ .                                |
| Inter-round drift                               | $\xi_t, \Delta_T, [a]_+$                                                           | Inter-round residual, adverse-drift budget, and positive part.                                                         |
| Visited set and radius                          | $\mathcal{K}_k^t, R_k^t$                                                           | Evaluated round- $t$ points and their maximum restricted distance from $Z_k^t$ .                                       |
| Minimum coordinate weight                       | $\Omega_k^{t,\min}$                                                                | Minimum $\min_{m \in \mathcal{S}_k^t} \Omega_m^t$ for a nonempty active neighborhood.                                  |
| Surrogate error                                 | $\delta_k^t$                                                                       | Two-sided exact-to-surrogate coupling error.                                                                           |
| Majorization slack                              | $\rho_k^t, \rho_k^{t,s}$                                                           | Per-update lift slack and its local-step value at $s$ .                                                                |
| Participation gap                               | $\beta_k^t$                                                                        | Squared gradient gap between full and participating exact blocks.                                                      |
| Simultaneous-update residual                    | $\chi^t, \Xi_T$                                                                    | Per-round interaction residual and cumulative positive-part budget.                                                    |
| Gradient consistency                            | $\kappa$                                                                           | Constant linking the surrogate/exact gradient gap to $\delta_k^t$ .                                                    |
| Averaged gaps                                   | $\bar{\delta}_T, \bar{\beta}_T$                                                    | Time-averaged surrogate and participation gaps.                                                                        |
| <b>Appendix and experimental notation</b>       |                                                                                    |                                                                                                                        |
| Bayes analysis                                  | $\mathcal{G}_k, f_k^*$                                                             | Observation $\sigma$ -field $\mathcal{G}_k = \sigma(X_{\mathcal{M}_k})$ and Bayes predictor.                           |
| Heterogeneity                                   | $\alpha$                                                                           | Dirichlet concentration controlling client distribution skew.                                                          |
| Metrics                                         | ACC, UAR, F1                                                                       | Accuracy, unweighted average recall, and F1 score.                                                                     |